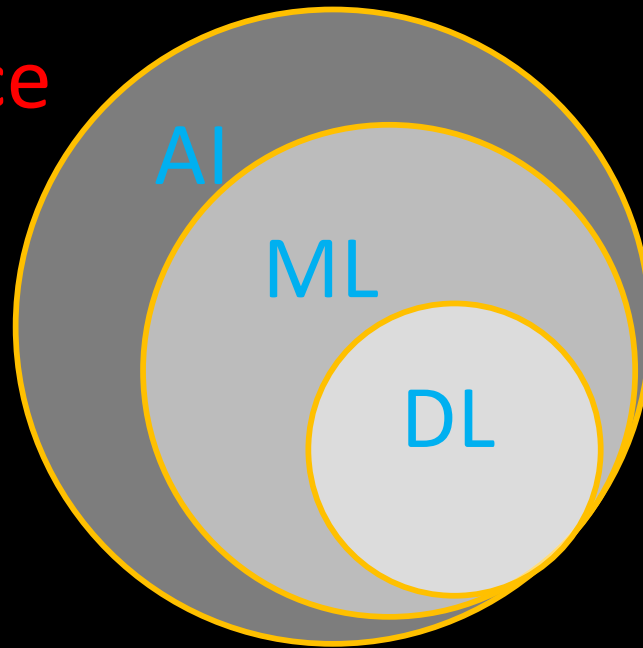


AI (Goal):
Artificial intelligence



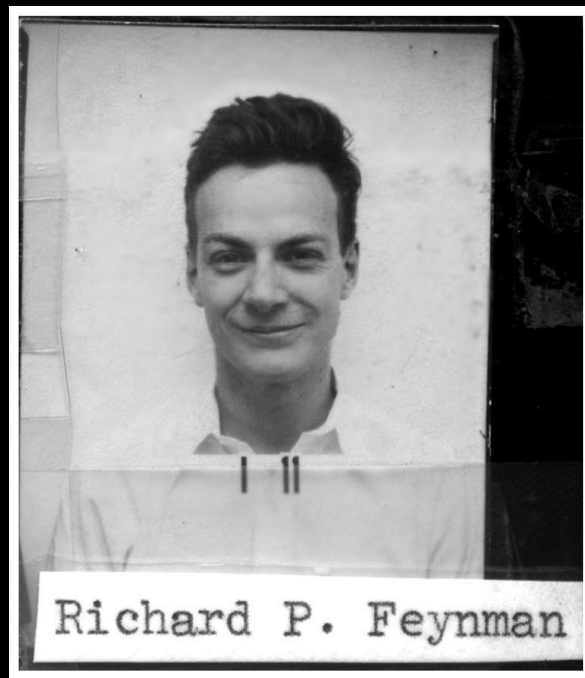
ML:
machine learning

DL:
deep learning

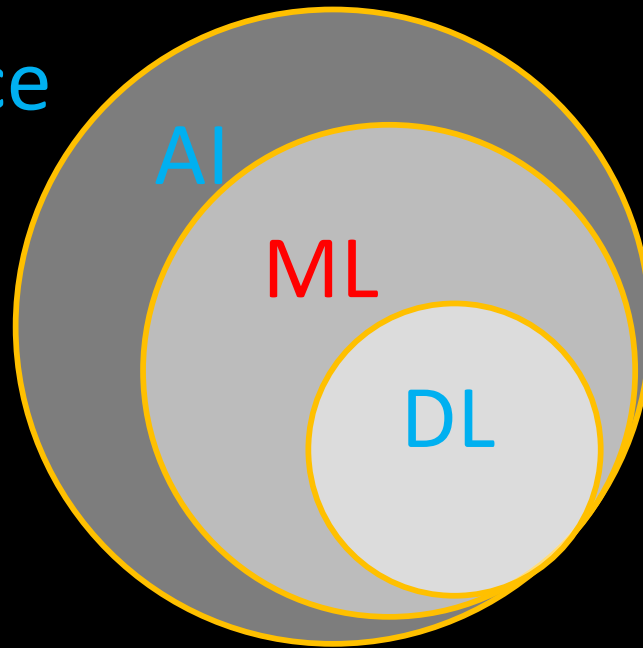
[github.com/changlabntu
/ML2025](https://github.com/changlabntu/ML2025)



1. “human-level” intelligence has a diverse definition
2. AI may reach human level intelligence, but probably are not doing it in a “humane” way



AI (Goal):
Artificial intelligence



ML:
machine learning

DL:
deep learning

ML: machine learning

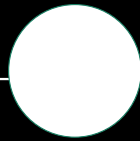
(collection of methods) to develop
algorithms automatically by learning
from data

Option 1: Manual-Crafted Rule Based



Wheels > 4

Truck



Height > 2m

Truck



Color = red

Truck?

Truck = (Wheels > 4)
and / or
A Good Rule
(Height > 2m)

Truck = (Color == red)
A Bad Rule

Hand crafted (explicit rules)



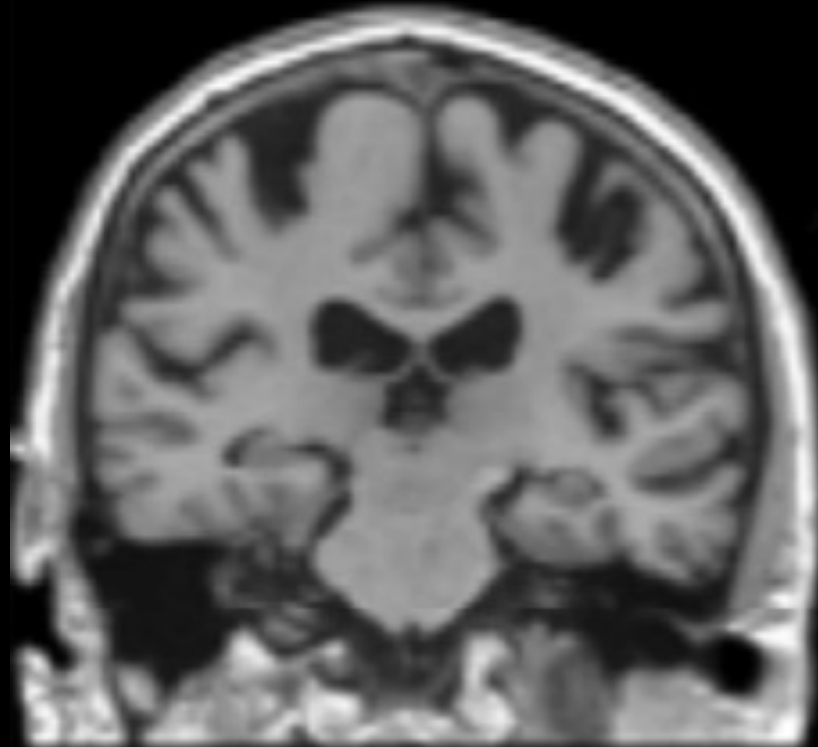
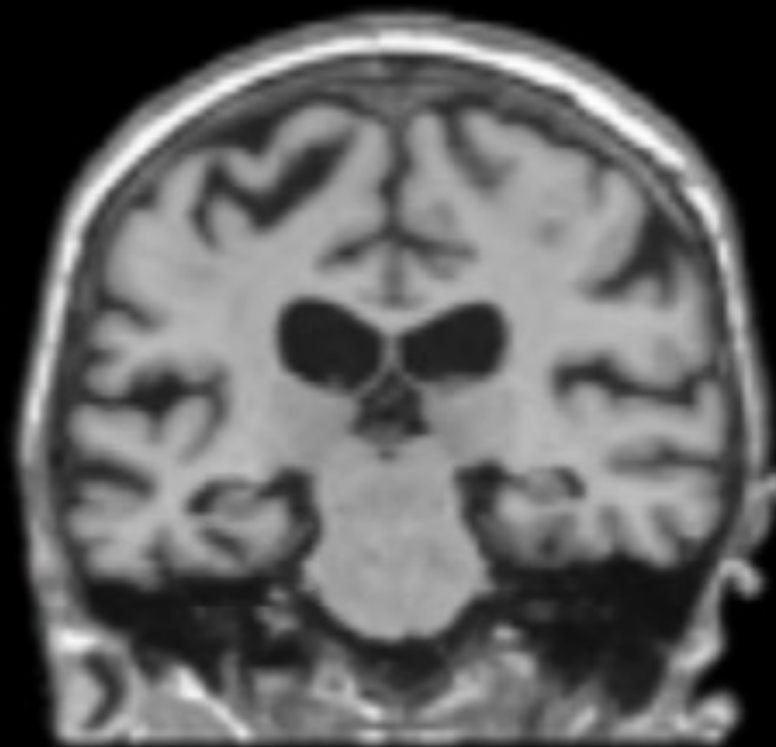
How to do define a function
to detect tires?

get_circle

get_circle + is_tire
(black rims larger than....)



get_circle + is_tire +
tires_belong_to_the_same_car....



Complicated, real-life problems can't simply
be resolved by rules

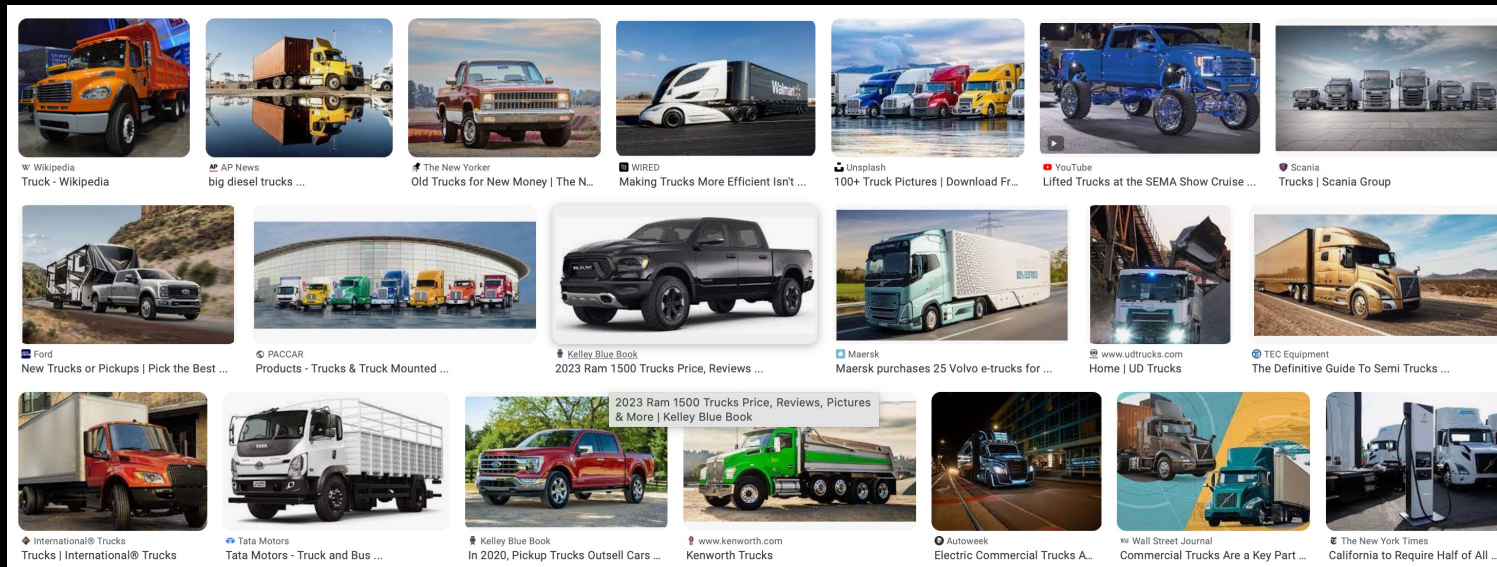
Option 2: let the model to learn from data: Machine Learning

 f

INPUT

)

OUTPUT



Model + Data + ways for the model to improve itself

Machine learning:

Learning a model that don't have explicit rules,
instead the model is derived implicitly from
observations (data)

when,

Rules are hard to be implemented
&

We don't know the rules clearly.

Look for Function(**s**)

f (INPUT) $\longrightarrow$ OUTPUT

Visuals, audios, text...

Labels, predictions,
decisions...

Look for Function(**s**)

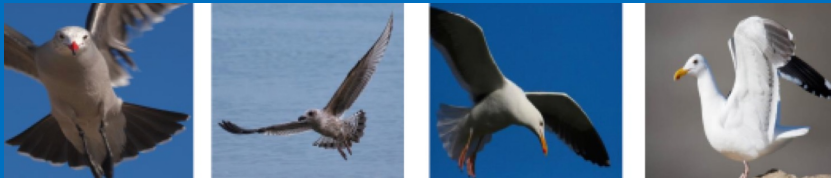
f (INPUT) $\longrightarrow$ OUTPUT
“Magic”

(turn out it's neural networks)

Machine Learning: Look for Function(**s**)

Classification

$f($



)



Class
(what kind of bird)

Detection

$f($



)



Class (people lamp),
Box location,
Box size

Segmentation

$f($

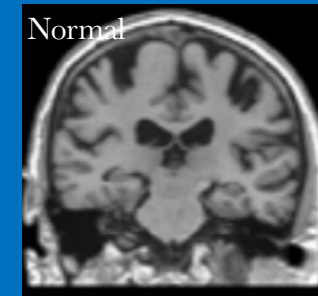
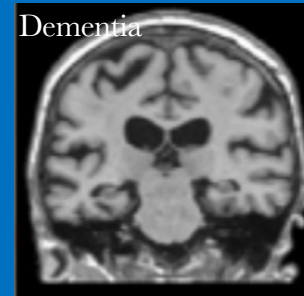


)



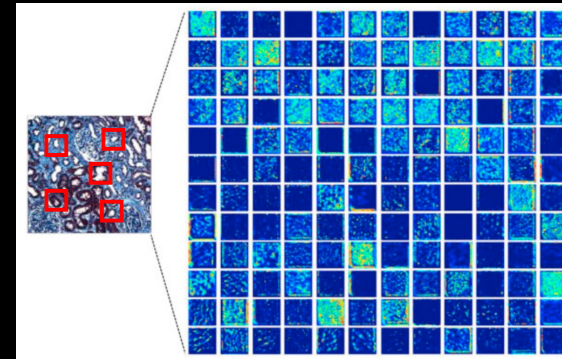
Deep Learning in Medicine

Classification



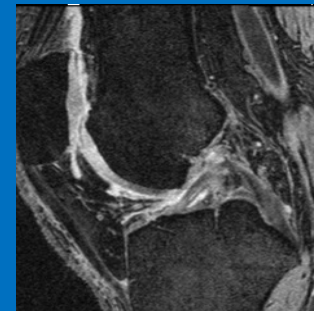
Dementia
Prediction

Detection



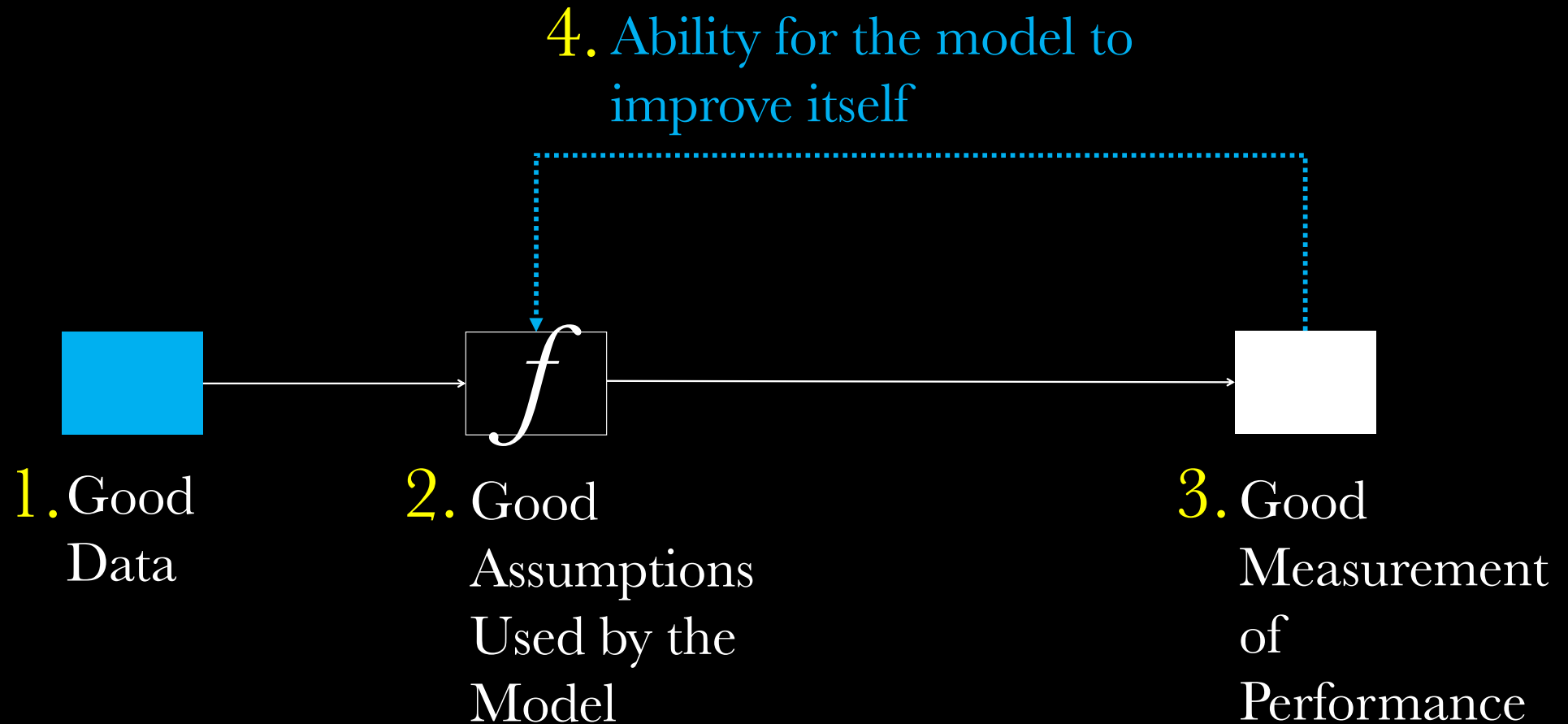
Hypocellularity
Detection

Segmentation

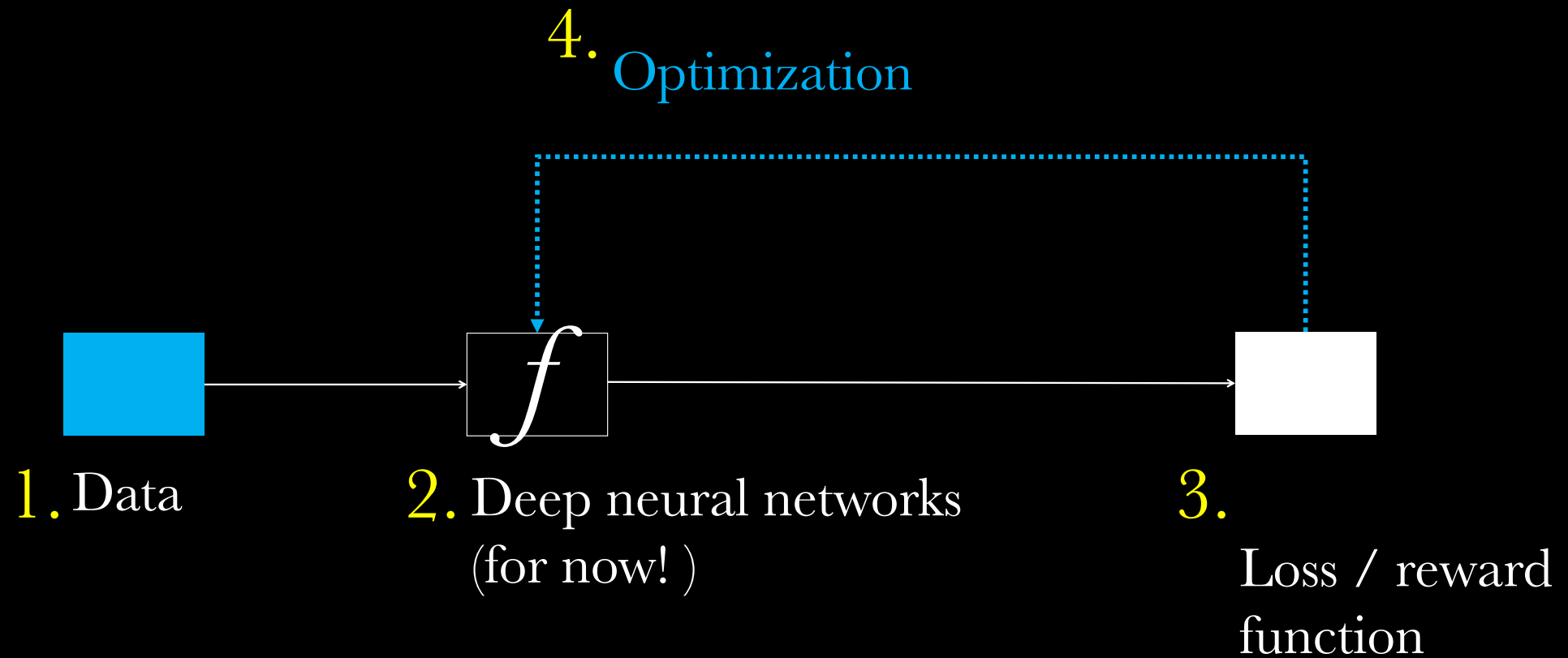


knee anatomical
segmentation

Four Horseman of ML



Four Horseman of ML

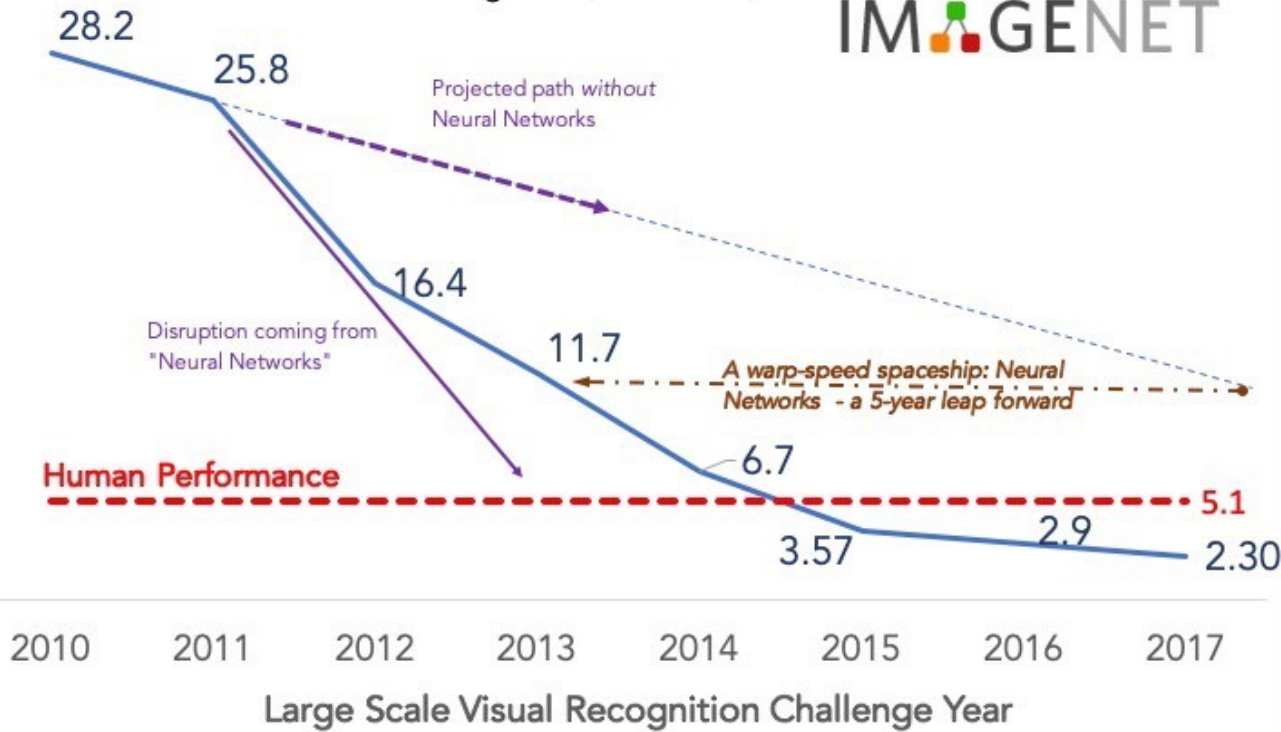


Accuracy of object detection software against the human eye

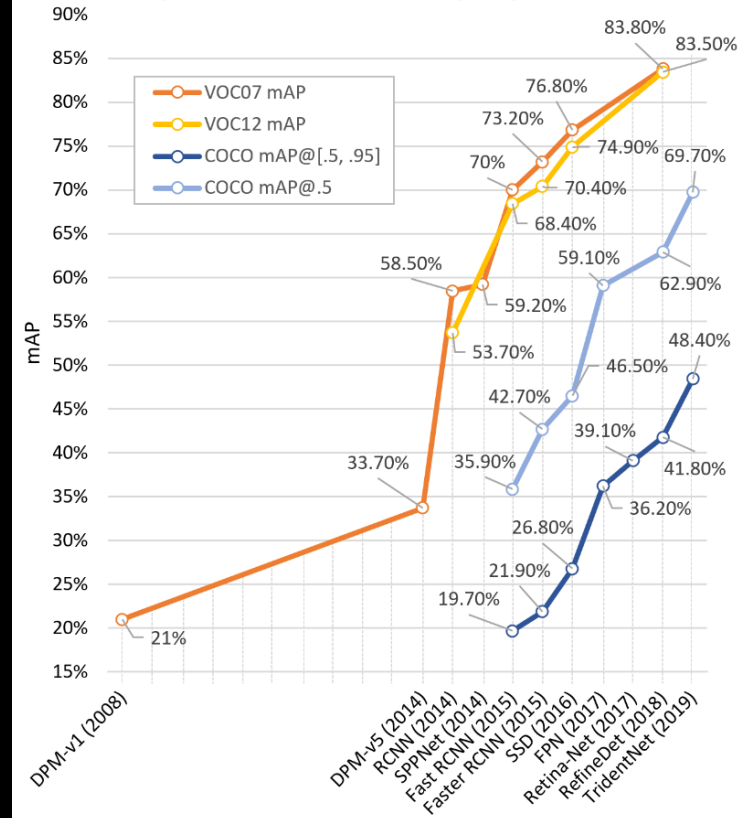
ImageNet (2010-2018)

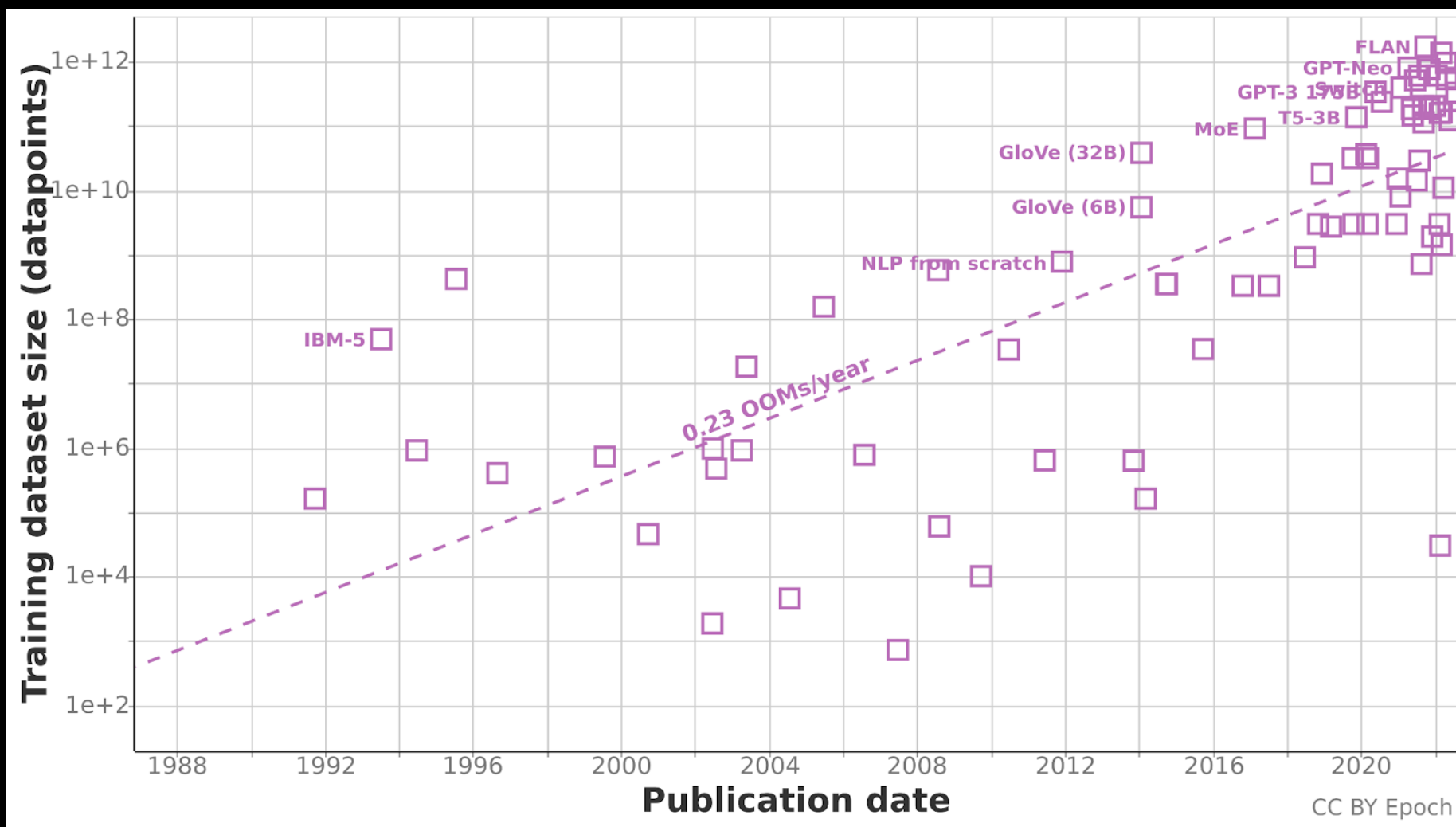


Percentage Error Rate %



Object detection accuracy improvements





Three Categories of AI Application in Medicine (or anything!)

Human
Can't

AI
Can



Human
Can but Expensive

AI
Equal quality but Automatic

Human
Can but unavailable

AI
Equal or Less but acceptable

Most Simple ML:

Linear & Logistic Regression

Regression vs Classification

Regression:

Predicts **continuous outcomes** by finding the best-fit line through data points.

Classification:

Predicts the **probability of an instance** belonging to a particular class.

Regression:

Predicting drug dosage $\leftrightarrow$ patient weight

Estimating bone density changes $\leftrightarrow$ X-ray

blood pressure $\leftrightarrow$ salt intake

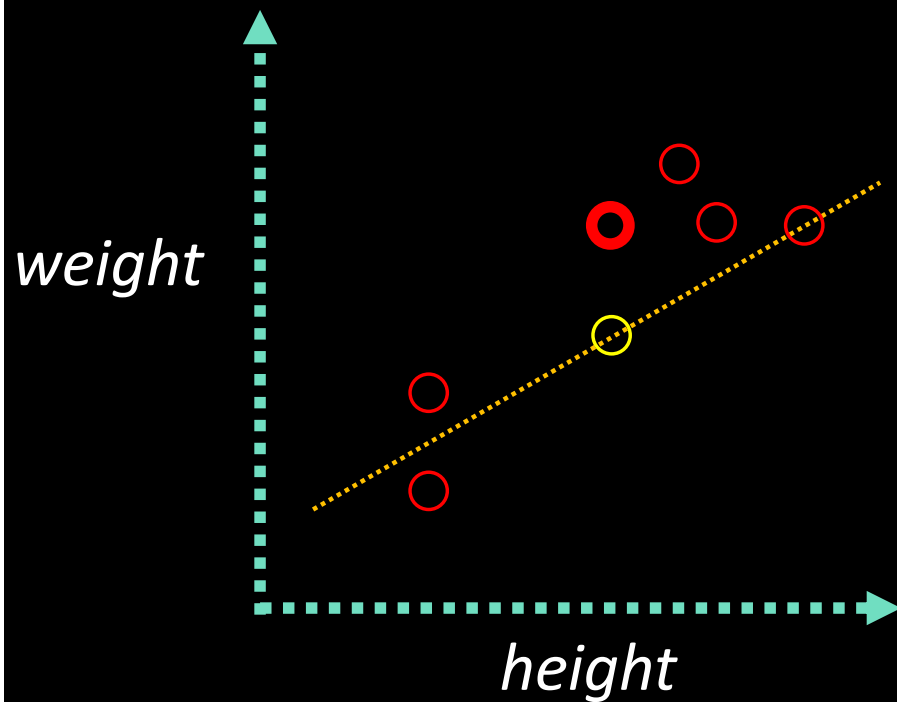
Classification:

likelihood of heart disease $\leftrightarrow$ cholesterol levels

probability of remission in patients $\leftrightarrow$ treatment

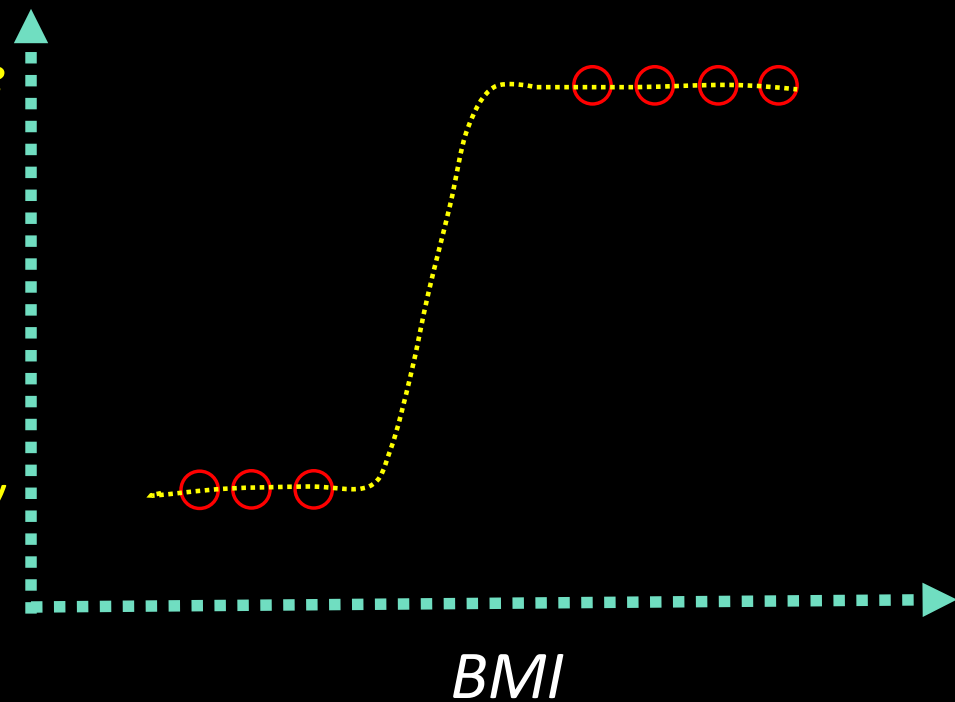
Classifying medical images $\leftrightarrow$ presence or absence of a tumor

Regression vs Classification



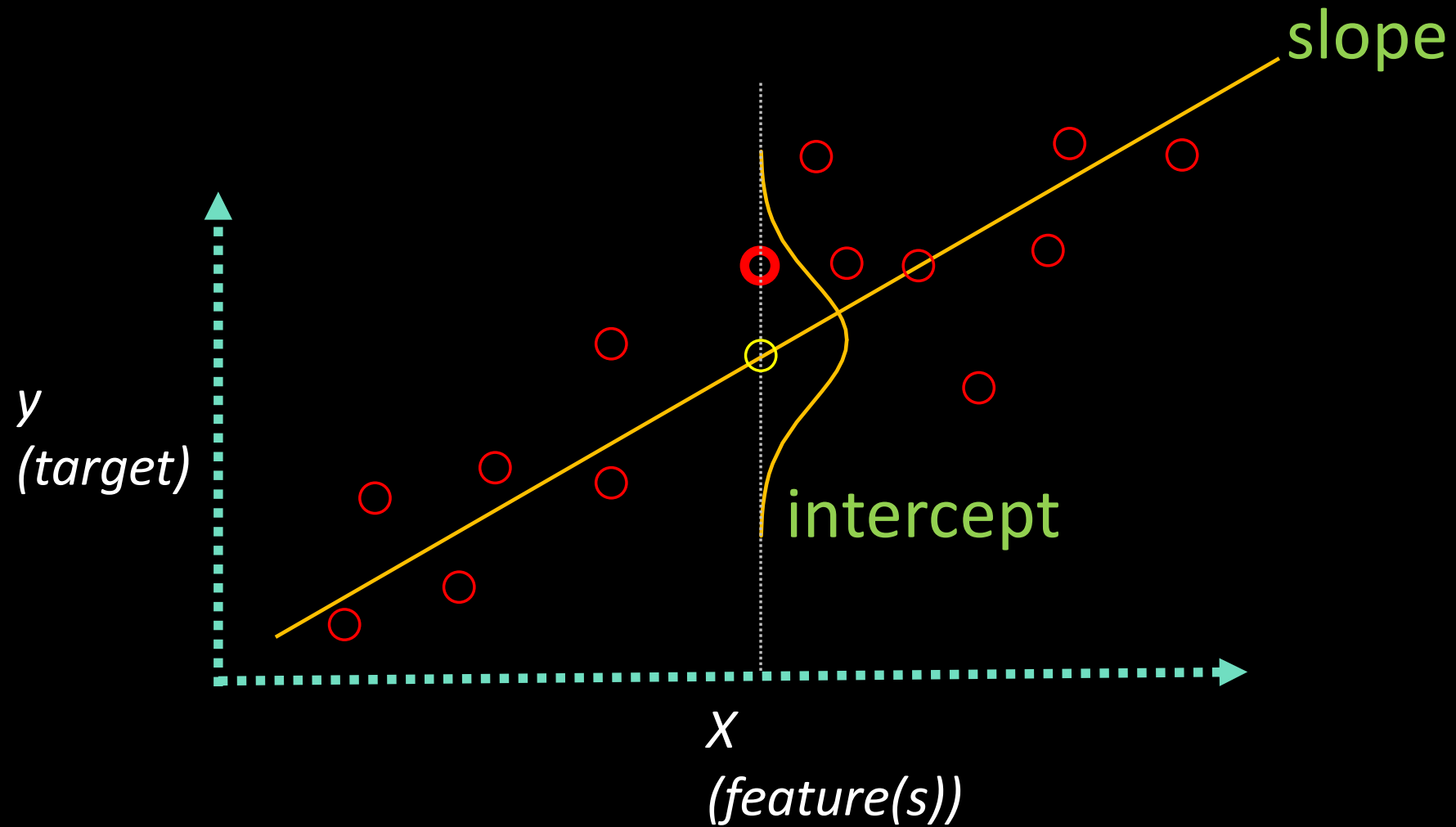
1=disease

0=healthy

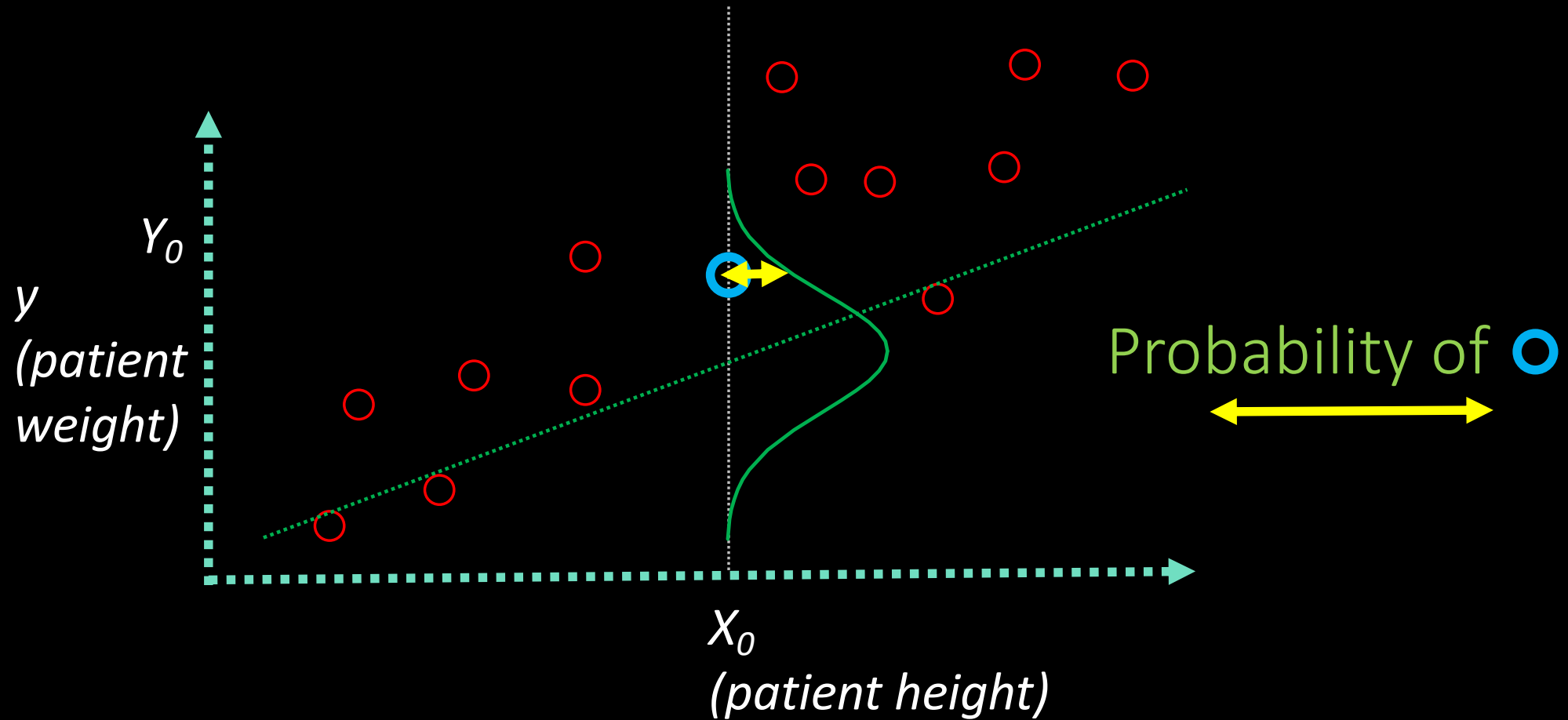


but why?

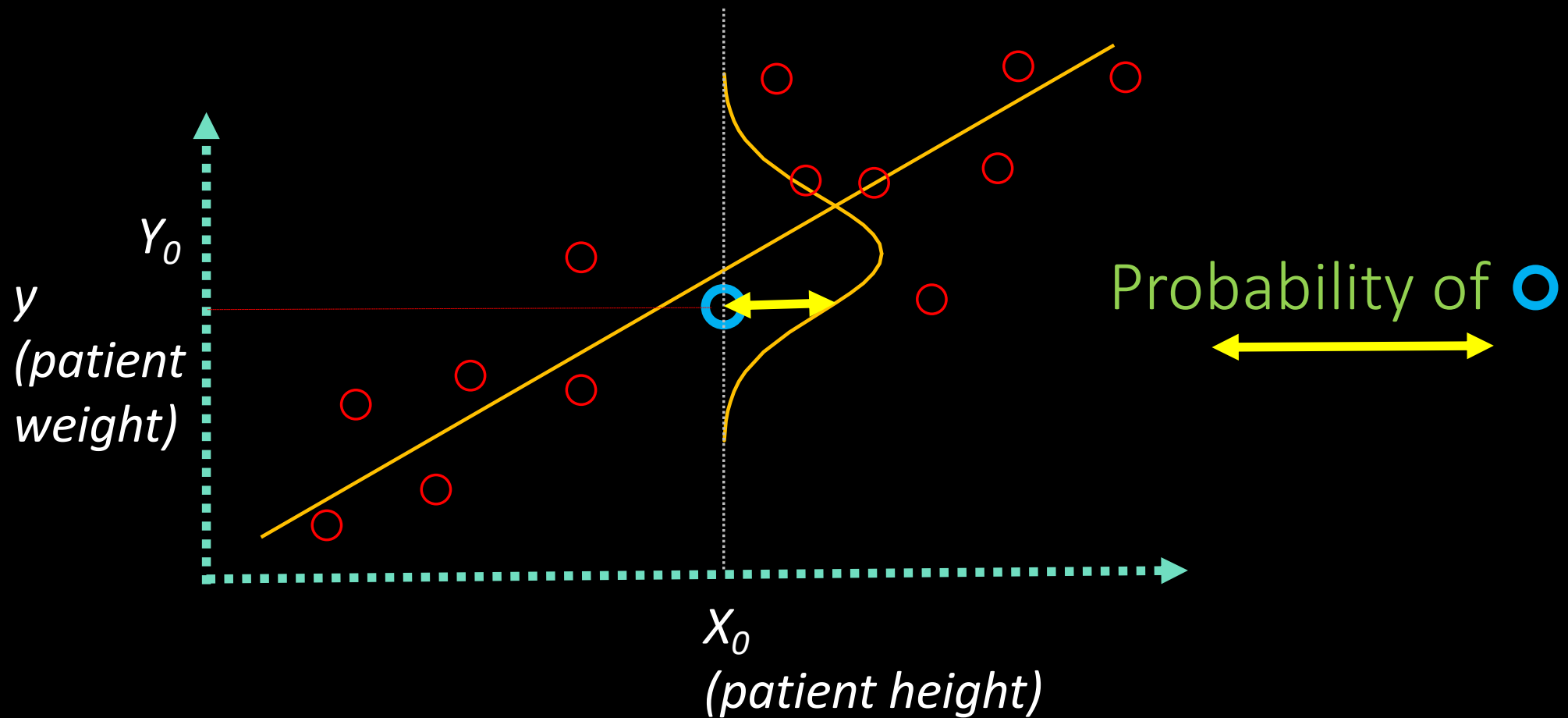
What are the parameters?



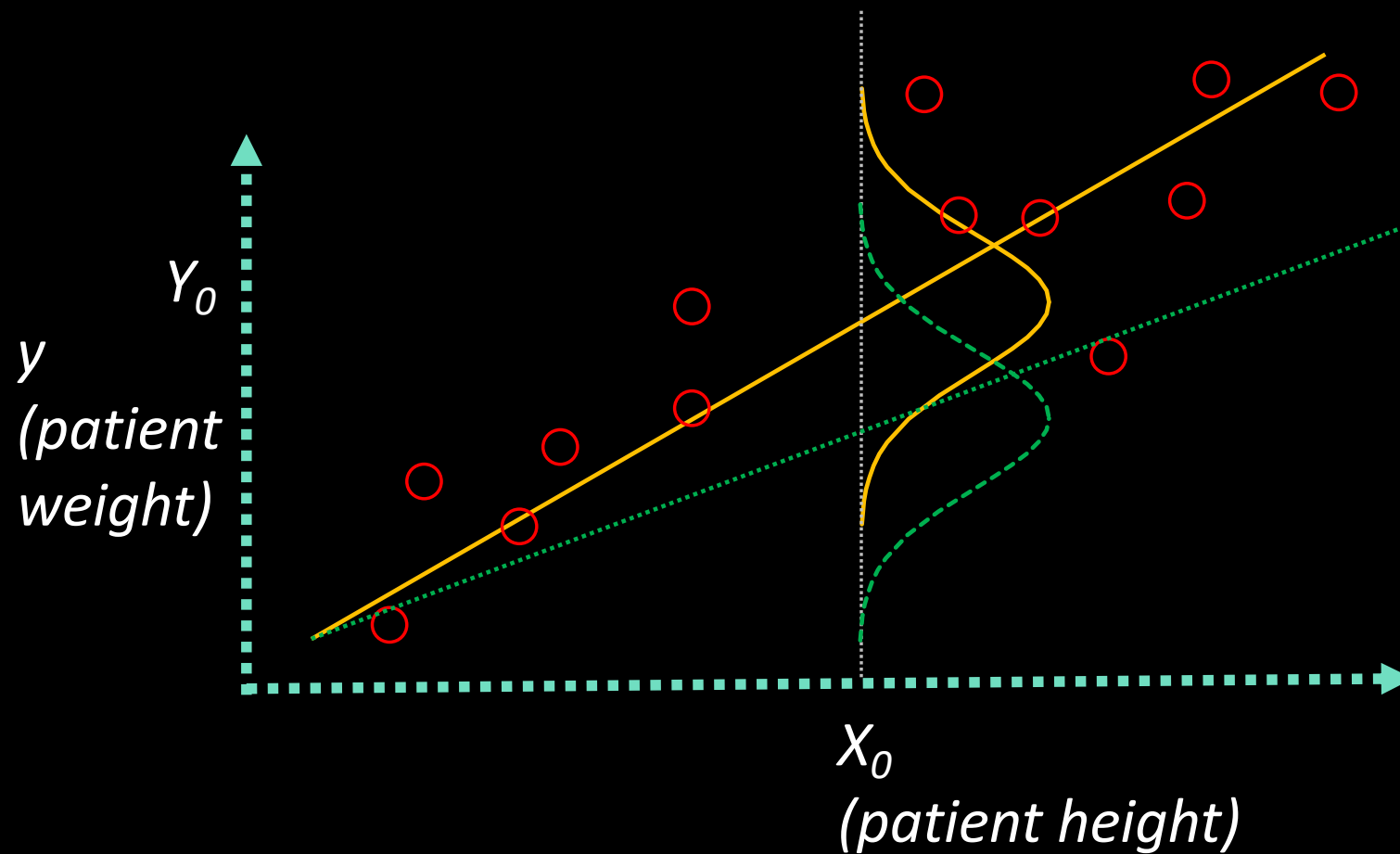
What's the less probable model?



What's the more probable model?



Find the more probable model:
Maximum likelihood estimation



Maximum Likelihood Estimation (MLE)

$$\theta_{ML} = \underset{\theta}{\operatorname{argmax}} P(y|X; \theta)$$

It finds the parameter values that make the observed data most probable.

Don't be scared by the math!

$$\theta_{ML} = \underset{\theta}{\operatorname{argmax}} P(y|X; \theta)$$

θ means parameters

$\operatorname{argmax}(*)$ Means adjusting the parameters until
(*) at its maximum

$P(y|X; \theta)$ Means the conditional probability:
 y appears *given* X

Maximum Likelihood Estimation (MLE)

In the foundational of many ML models,
including language models

$$\theta_{ML} = \underset{\theta}{\operatorname{argmax}} P(y|X; \theta) \quad y = M_{\theta}(X)$$

Output sentences

input sentences

Maximum Likelihood Estimation (MLE)

Training a polite chatbot:

$$\theta_{ML} = \underset{\theta}{\operatorname{argmax}} P(\text{"I am fine, thank you"} | \text{"How are you"}; \theta)$$

Training a passive aggressive chatbot:

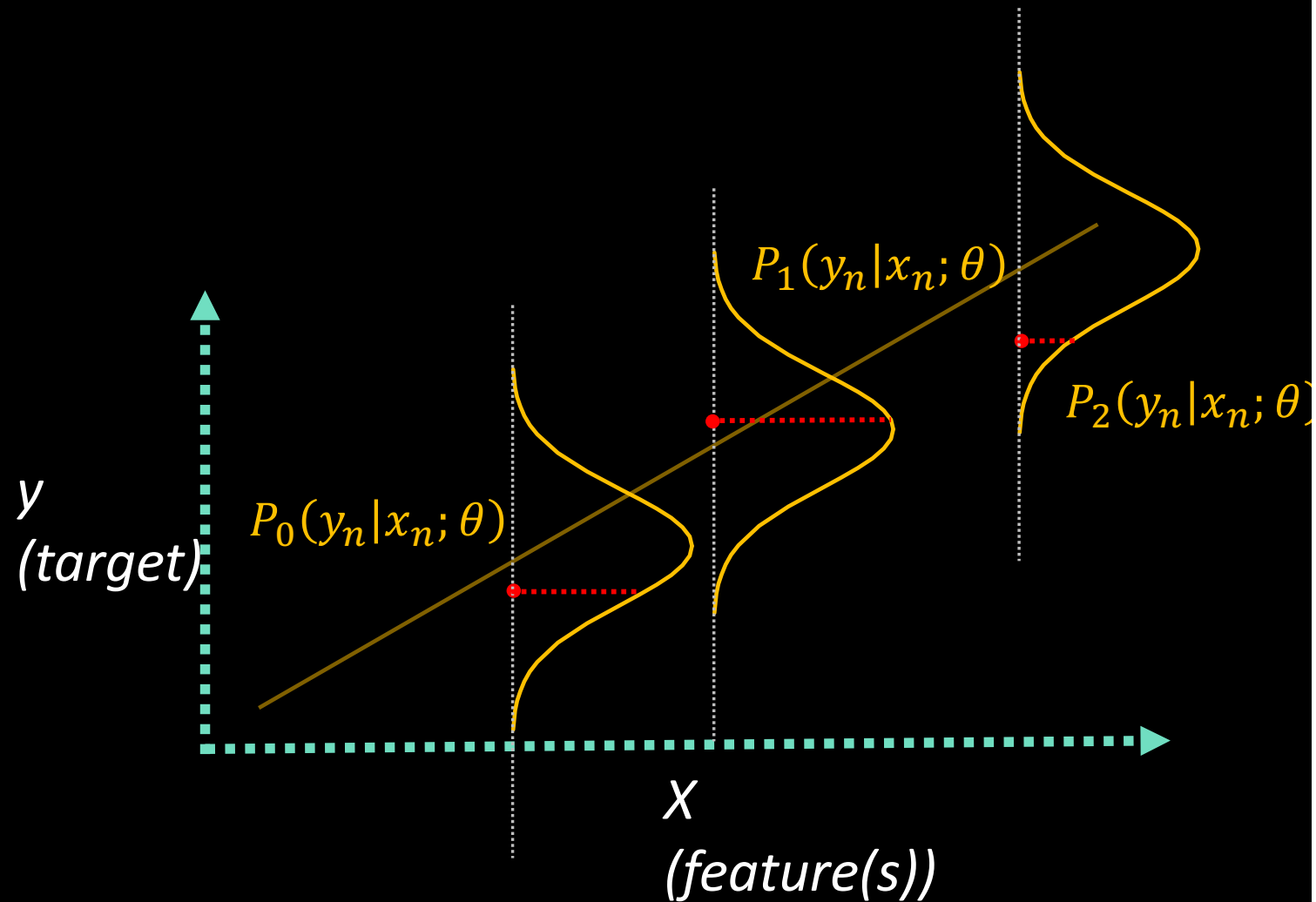
$$\theta_{ML} = \underset{\theta}{\operatorname{argmax}} P(\text{"Why do you care? "} | \text{"How are you"}; \theta)$$

How to evaluate
many data
points at once?

$$\prod_n P(y_n | x_n; \theta)$$

$$= P_0 * P_1 * P_2 \dots$$

Times them up
together



Datapoints 1. 2. 3 4. 5

X : what's the capital of France, USA, Japan, UK, Germany

Y: Paris DC Tokyo London Berlin

Probabilities according to the bad model:

[0.171 0.110 0.06 0.03 0.01.....

Probabilities according to the good model:

[0.498 0.598 0.398 0.598 0.698.....

Joint Probability

$$P(y_1|x_1; \theta) * P(y_2|x_2; \theta) \dots * P(y_n|x_n; \theta) = \prod_n P(y_n|x_n; \theta)$$

$$\prod_n P(y_n|x_n; \theta) \quad (0.1 * 0.1 * 0.1 * 0.1 \dots) \quad 0.0000000001$$

VS

$$\sum_i \log P(y_n|x_n; \theta) \quad (-1 -1 -1 -1 -1 \dots) \quad -9$$

Log Probability

If you want to see the derivation

$$\sum_n \log P(y_n | x_n; \theta)$$

Good Old Normal Distribution

$$= n \sum \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(y_n - wx_n - b)^2}{2\sigma^2} \right) \right)$$

$$= -n * \log \frac{1}{\sqrt{2\pi\sigma^2}} - \sum_i \frac{(y_n - wx_n - b)^2}{2\sigma^2}$$

Assuming constant variance

$$\operatorname{argmax}(\sum_i \log P(y_n | x_n; \theta)) = \operatorname{argmin}(\sum_n (y_n - wx_n - b)^2)$$

$$= \operatorname{argmin}(\sum_n (y_n - \hat{y})^2)$$

Mean Square Error (MSE)

Linear Regression is optimized by
minimize mean square error (MSE)

(but you actually just need:)

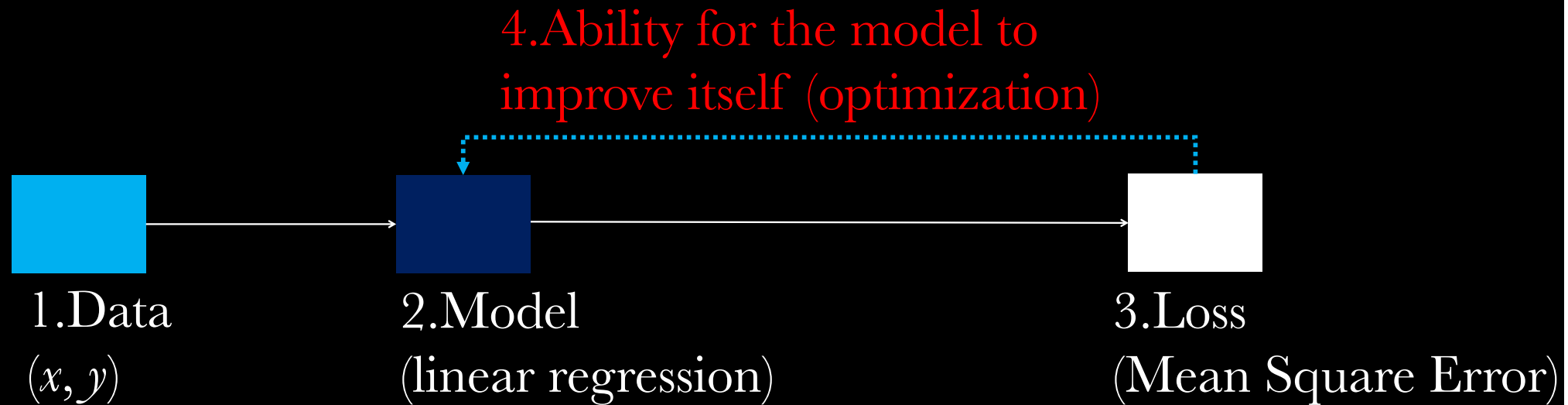
Regression by just called a library (Scipy)

```
from sklearn.linear_model import Regression
```

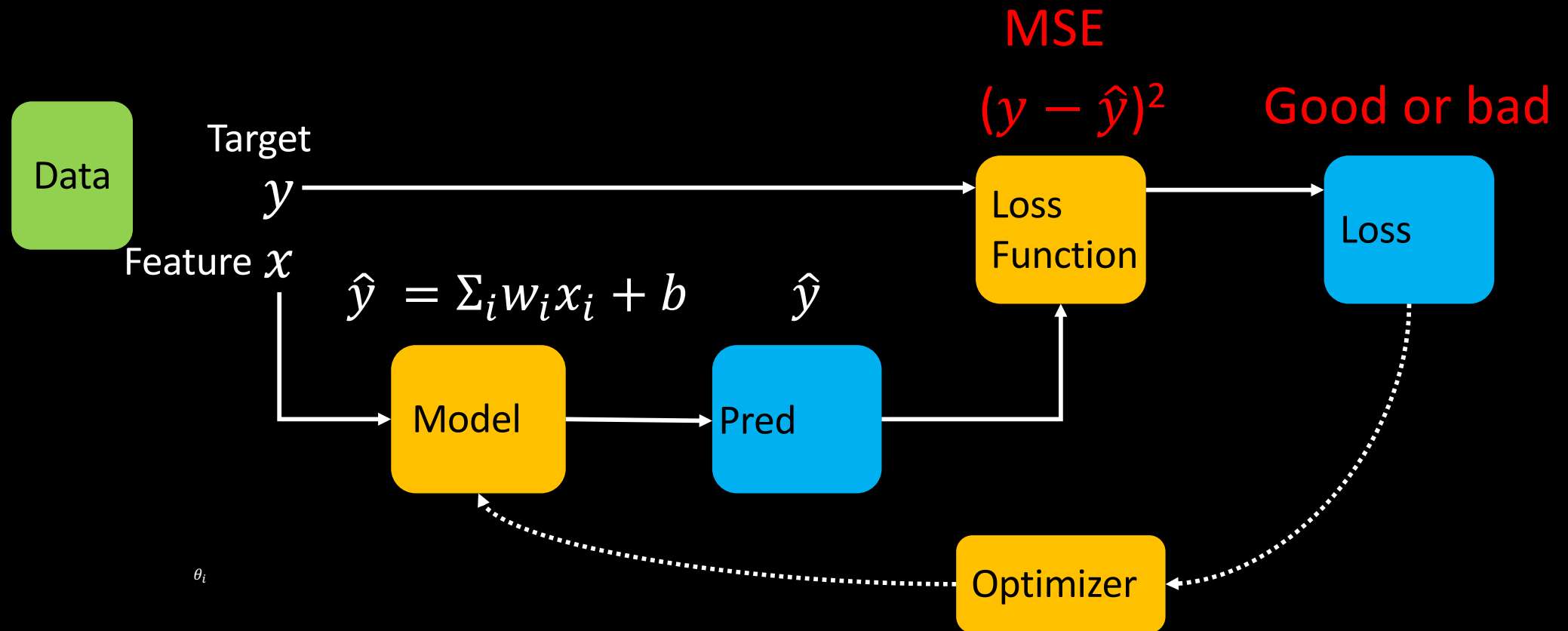
```
Regr = Regression()
```

```
Regr.fit(x_train, y_train)
```

Behind the Scene `Regr.fit(x_train, y_train)` Of linear regression

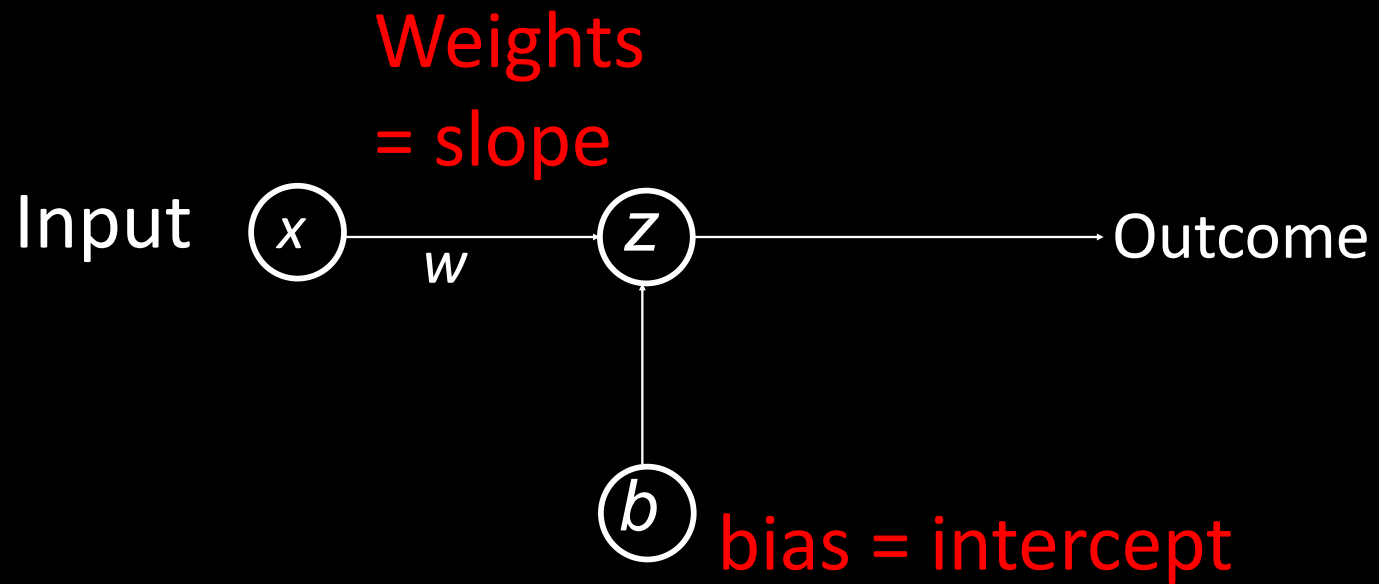


Flow Chart for Linear Regression



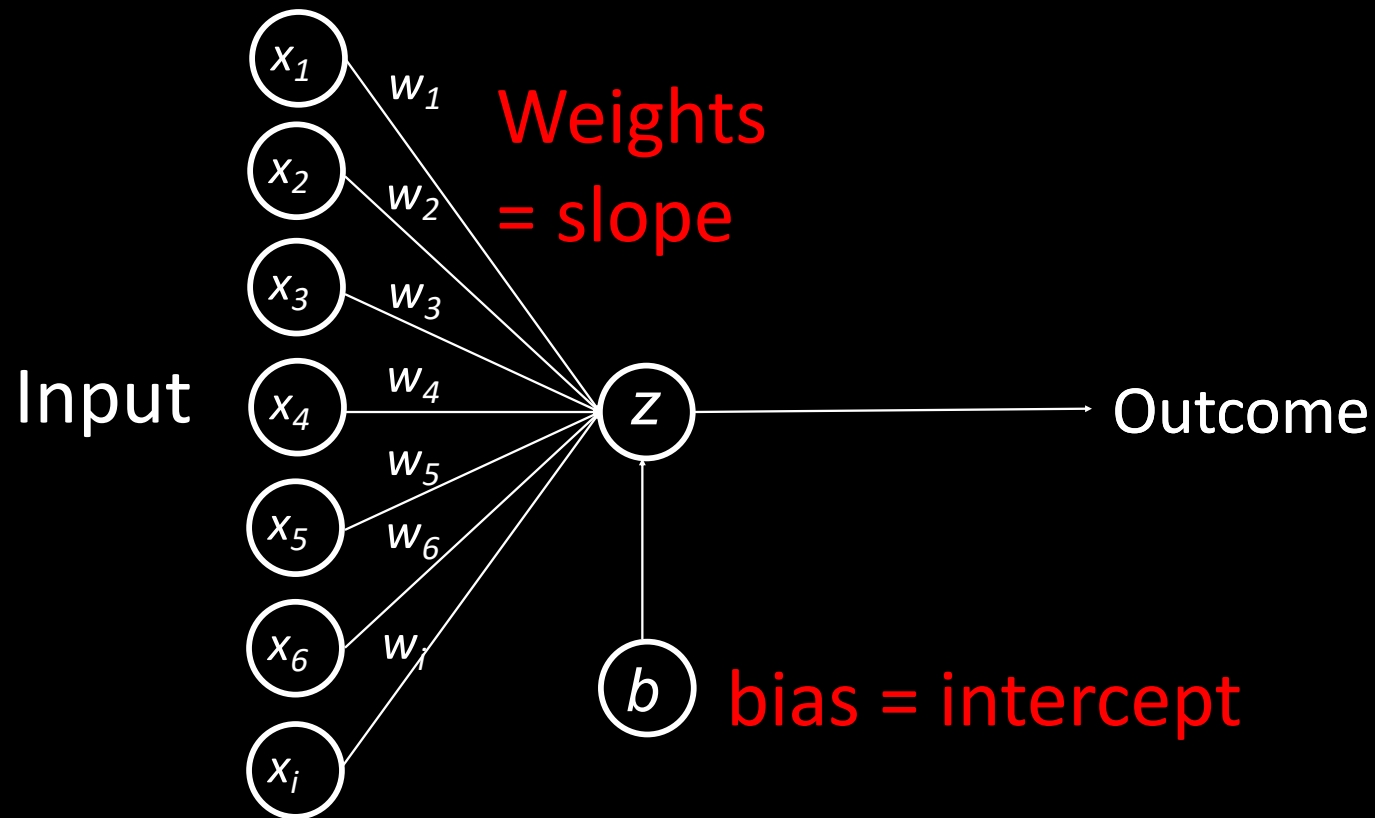
Graph of Linear Regression

$$y = WX + b$$

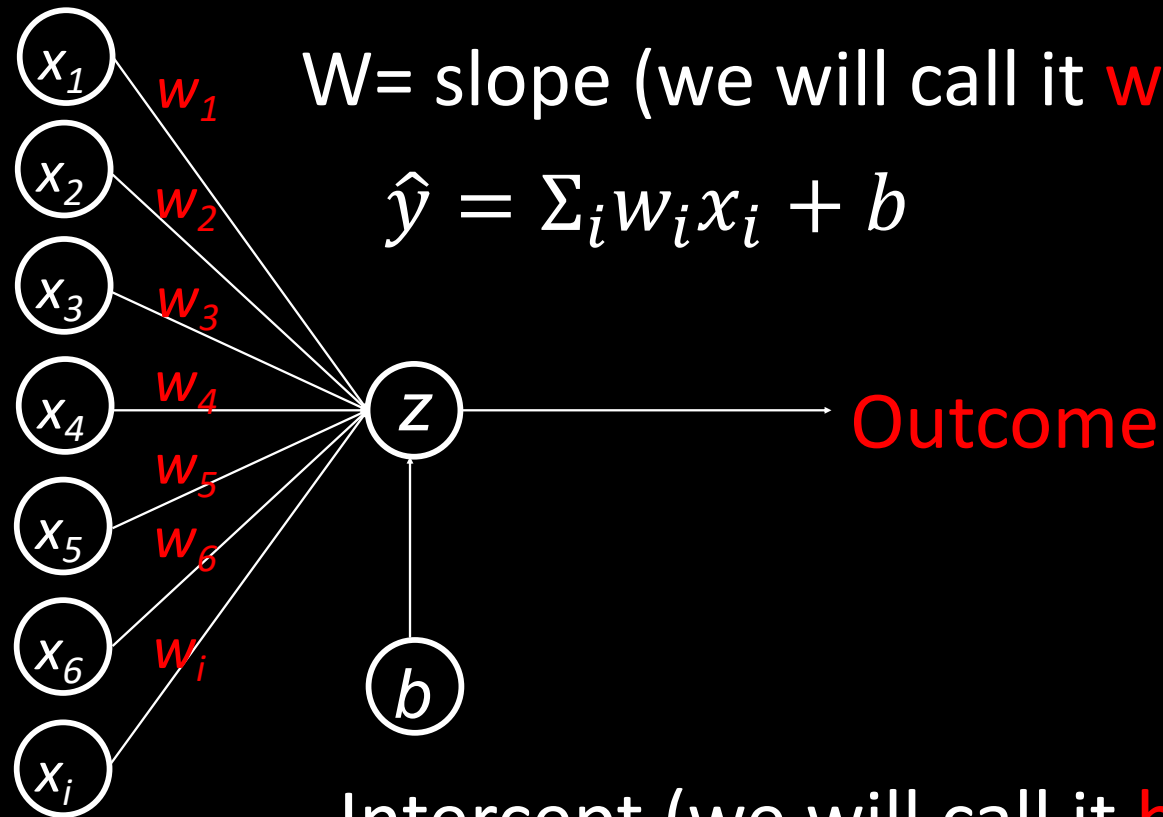


Graph of Linear Regression

$$y = w_1x_1 + w_2x_2 + \dots + b$$



If we re-draw it into a graph.....



Intercept (we will call it **bias** by some point)

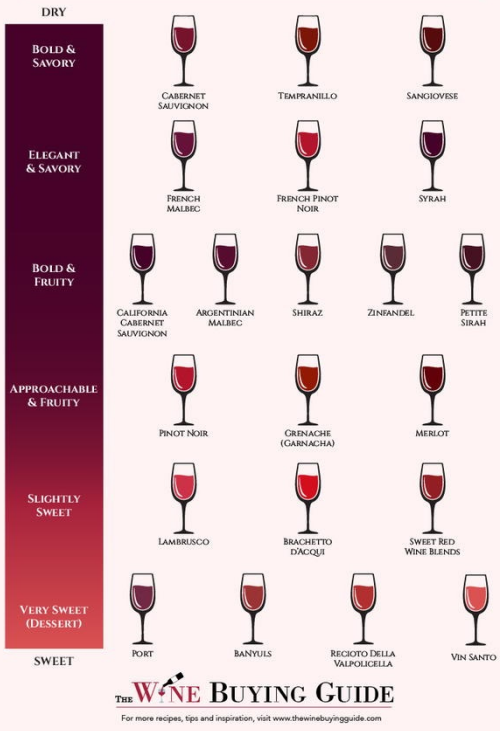
Classification: Logistic Regression

Human has a tendency (or need) to **categorize** things for our own perception

Table 2. Examples of the training data set (values are normalized between 0 and 1)

Fruit image	Input of the network		Output Fruit Grading
	Mean	Variance	
	0.43	0.11	'Extra'
	0.22	0.05	'Type 1'
	0.27	0.23	'Type 2'
	0.26	0.31	'Rejected'
	0.28	0.29	'Rejected'

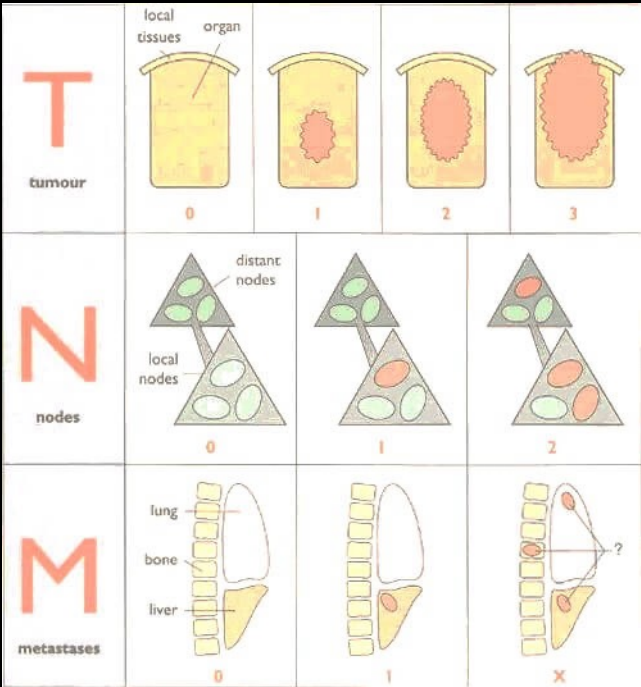
RED WINE SWEETNESS CHART



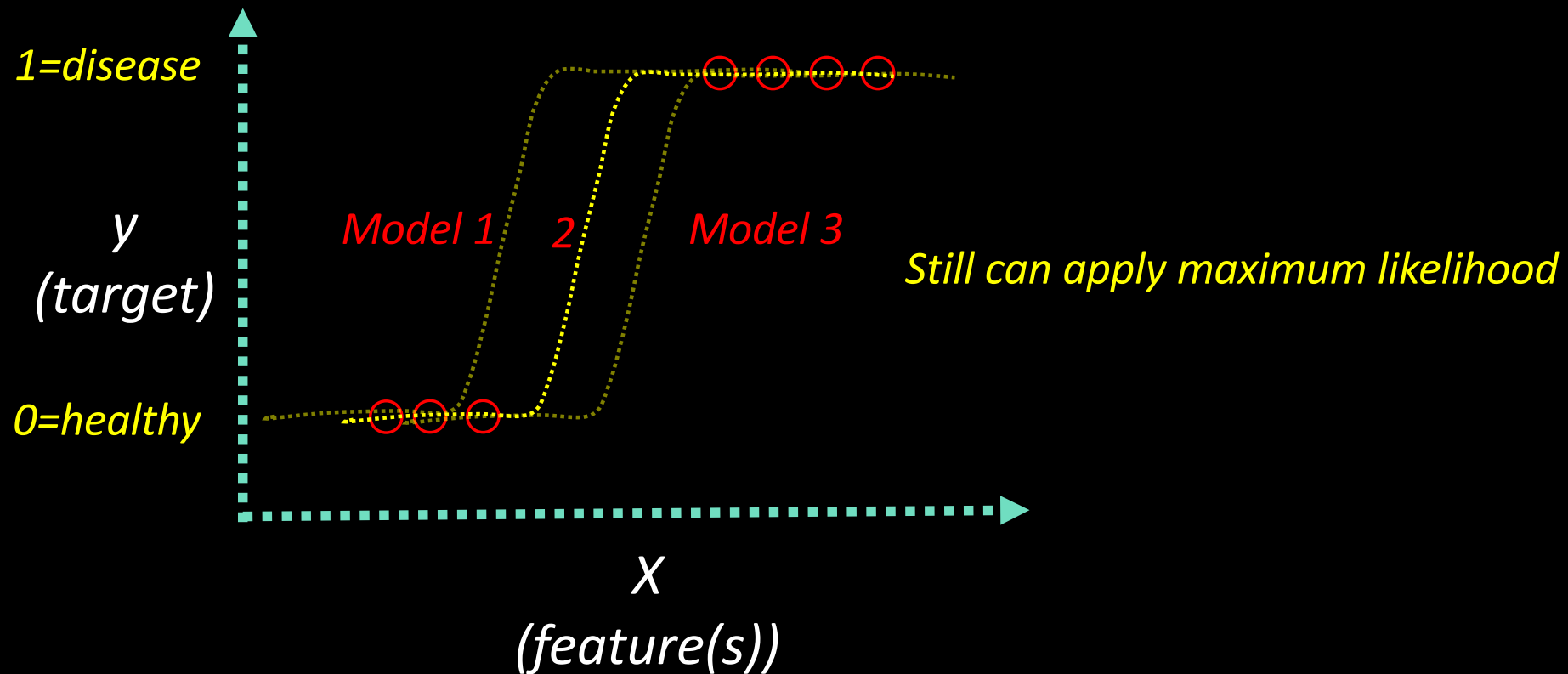
2023 Federal Tax Brackets

Tax Bracket/Rate	Single	Married Filing Jointly	Head of Household
10%	\$0 - \$11,000	\$0 - \$22,000	\$0 - \$15,700
12%	\$11,001 - \$44,725	\$22,001 - \$89,450	\$15,701 - \$59,850
22%	\$44,726 - \$95,375	\$89,451 - \$190,750	\$59,851 - \$95,350
24%	\$95,376 - \$182,100	\$190,751 - \$364,200	\$95,351 - \$182,100
32%	\$182,101 - \$231,250	\$364,201 - \$462,500	\$182,101 - \$231,250
35%	\$231,251 - \$578,125	\$462,501 - \$693,750	\$231,251 - \$578,100
37%	\$578,126+	\$693,751+	\$578,101+

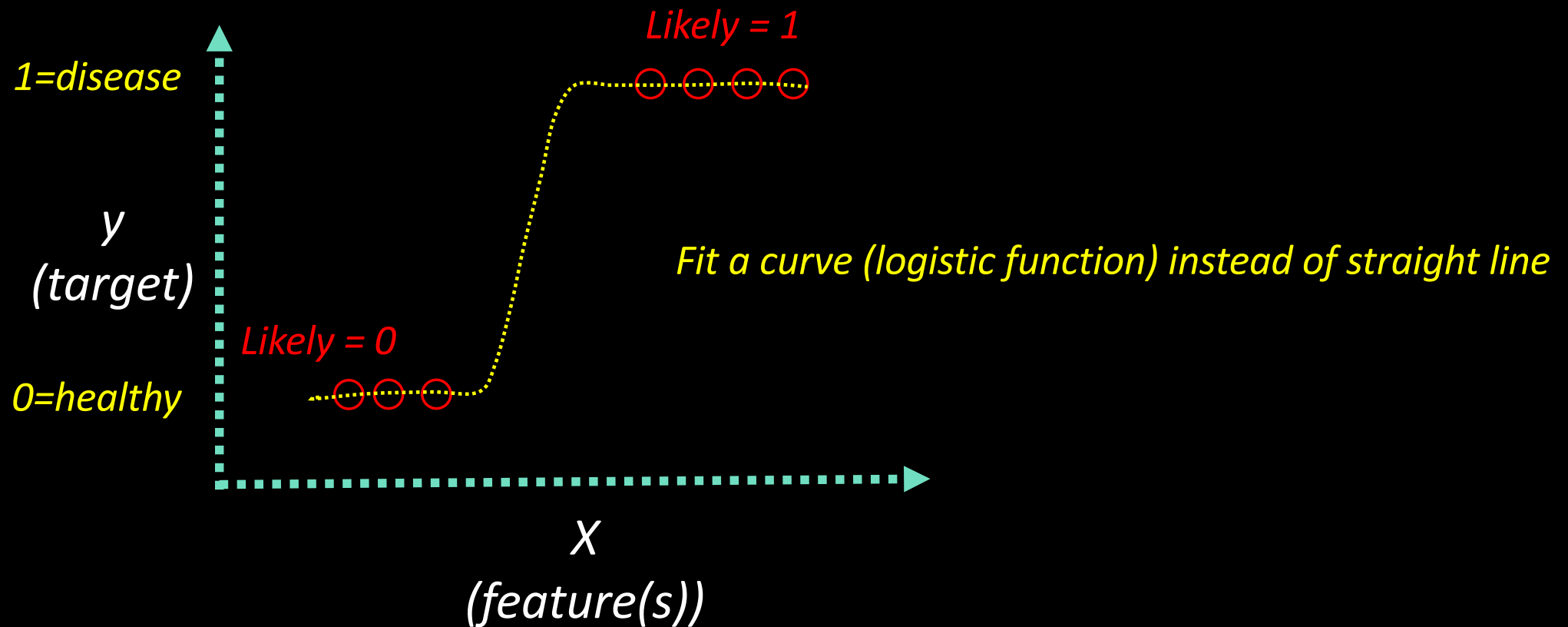
Source: TheCollegeInvestor.com



Logistic Regression: Predicts categorical outcomes



Logistic Regression: Predicts categorical outcomes



Chance of wolf (=1)



0 (very sure)



0.2 (not sure)



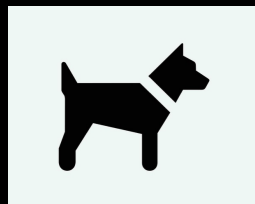
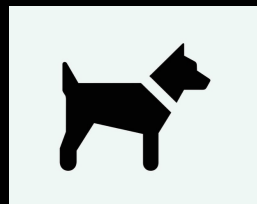
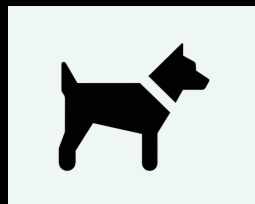
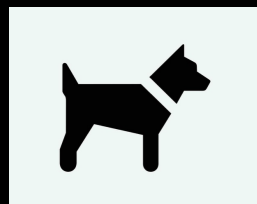
0.7 (not sure)



1 (very sure)



Clean cut



Clean cut

ambiguous

y
(Output Probability)

$P=0.5$

$P=0$

$P=0.2$

$P=0.7$

$P=1.0$



x
feature(s)

y
(Output Probability)



$P=0.5$

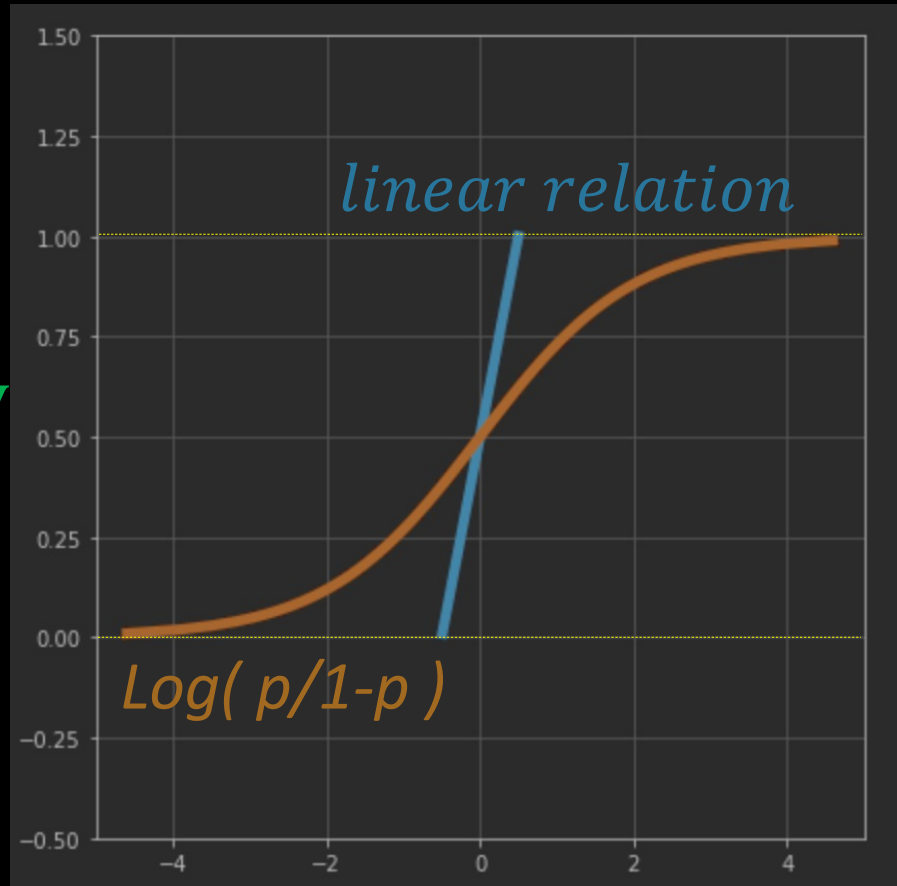
$P \geq 0.5$ *Class=1*

$P < 0.5$ *Class=0*

x
feature(s)



Probability
0-1



feature — $-\infty \sim \infty$

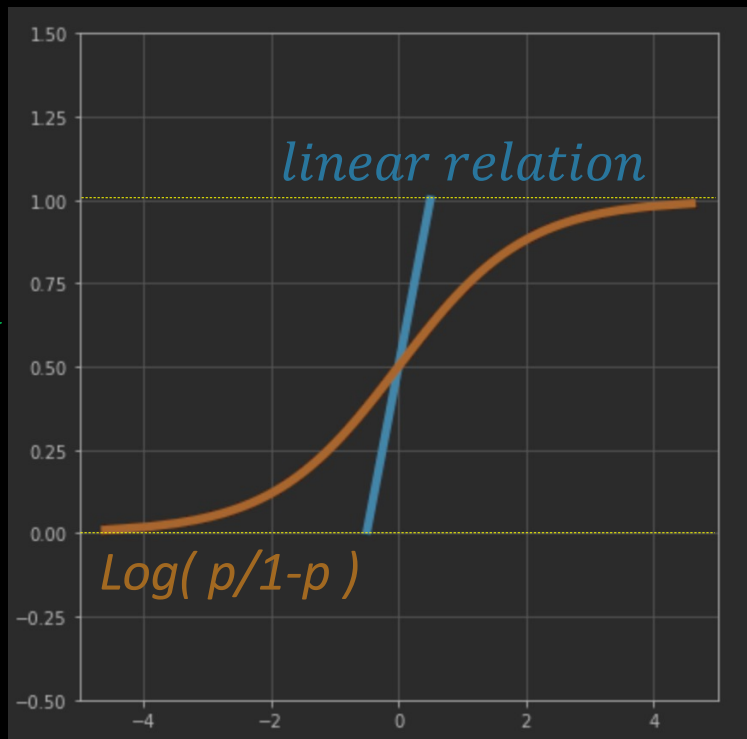
How to predict probability?

Remember we got
output of range

— $-\infty \sim \infty$

From regression

Probability
0-1



feature - ∞~∞

Probability p 0~1

Odd Ratio $\frac{p}{1-p}$ 0~∞

Logit $\log\left(\frac{p}{1-p}\right)$ -∞~∞

same range!

linear regression $\sum_i w_i x_i + b$ -∞~∞

Now we can just do regression to logit!

	No. of case patients	No. of control subjects	Odds ratio	95% confidence interval	<i>P</i> for trend*
Nonsmokers	117	1750	1.0	Referent	
Duration of tobacco use, y					
0.1–20.0	3	33	1.3	0.4–4.5	
20.1–32.0	7	33	3.4	1.4–8.0	
32.1–44.0	21	33	13.3	7.2–24.9	
≥44.1	30	30	19.1	10.4–35.1	<.0001
Average consumption of tobacco, g/day					
0.1–3.5	2	10	2.2	0.5–10.4	
3.6–5.0	22	54	7.9	4.3–14.3	
5.1–10.7	6	18	4.8	1.9–12.6	
≥10.8	31	47	12.4	7.2–21.4	.1
Cumulative consumption of tobacco, g/day × y					
1–71	3	31	1.3	0.4–4.4	
72–157	13	34	7.6	3.8–15.4	
158–382	15	32	8.0	4.0–15.7	
≥383	30	32	18.3	10.2–32.8	.0006
Age at start of tobacco use, y					
≤19	27	48	9.6	5.6–16.7	
20–26	20	52	6.3	3.5–11.2	
≥27	14	29	8.2	4.1–16.3	.4

*Test for linear trend, two-sided *P* value (considered statistically significant for *P*<.05).

Now we are mapping the
output of regression:

$$\Sigma_i w_i x_i + b \quad -\infty \sim \infty \quad \text{to}$$

To logit

$$\text{Logit} \log\left(\frac{p}{1-p}\right) \quad -\infty \sim \infty$$

$$p = \frac{1}{1 + \exp(-(\Sigma_i w_i x_i + b))} = \text{Sigmoid}(w_i x_i + b) = \text{S}(\beta X)$$

$$p = \frac{1}{1 + \exp - (\sum_i w_i x_i + b)} = \text{Sigmoid}(w_i x_i + b) = S(\beta X)$$

Just
Logistic
Regression

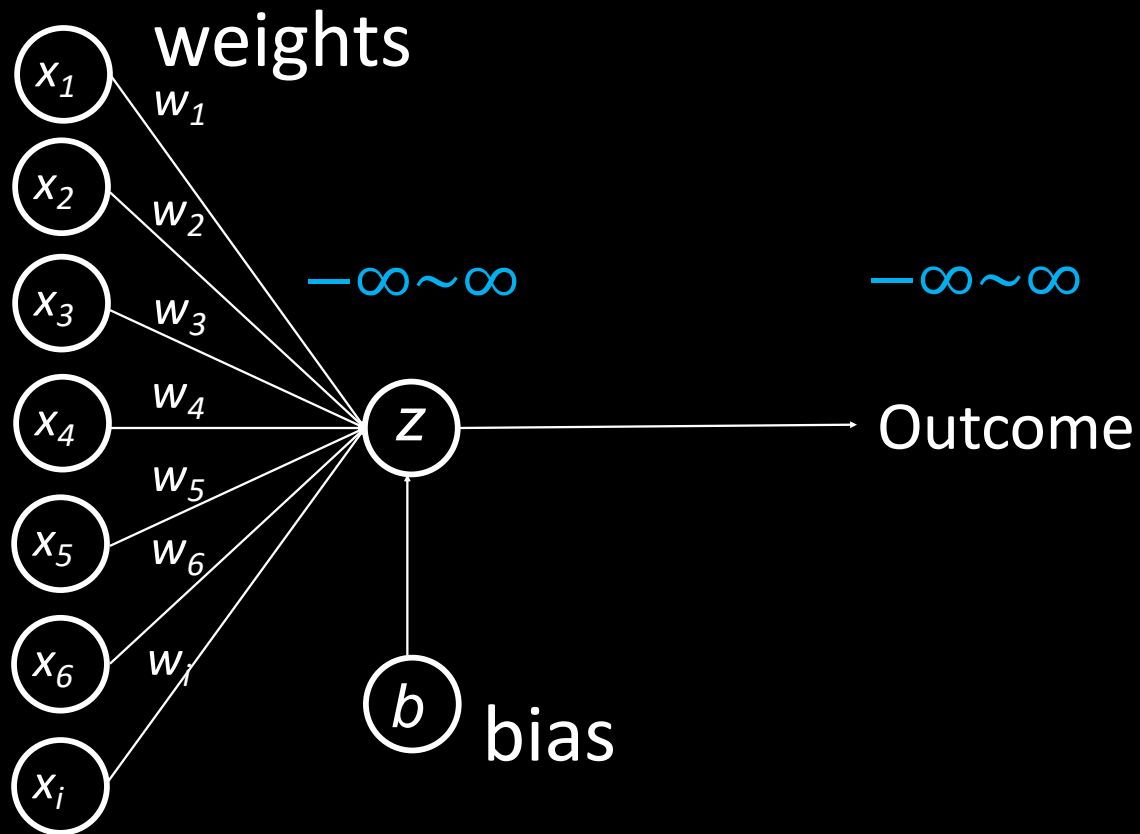
$$P = S(\beta X)$$

Logistic Regression = Regression
Between

Probability $\leftrightarrow$ Sigmoid of (weighted features)

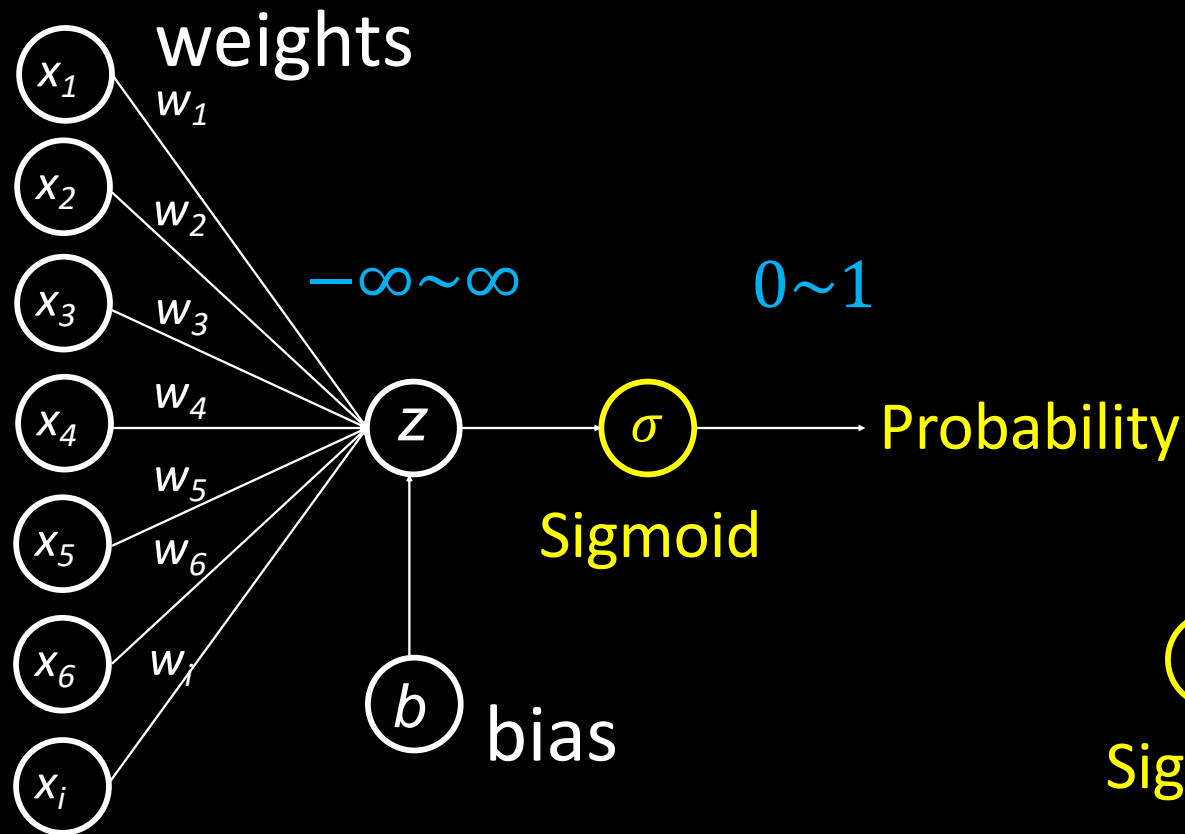
Graph of Linear Regression

$$y = (WX)$$



Graph of Logistic Regression

$$p = \frac{1}{1 + \exp - (\sum_i w_i x_i + b)} = \text{Sigmoid}(\sum_i w_i x_i + b) = S(WX)$$



Just add 1 more "layer"

σ
Sigmoid



Loss Function for Classification

(still need to way to say it's good or bad)

wolf

$P=1$

Probability
of
 $P=1$

$P=0$

dog



wolf



Wrong



Right



Right



dog



Wrong

wolf

$P=1$



Probability of $P=0$

Probability of $P=1$

$P=0$

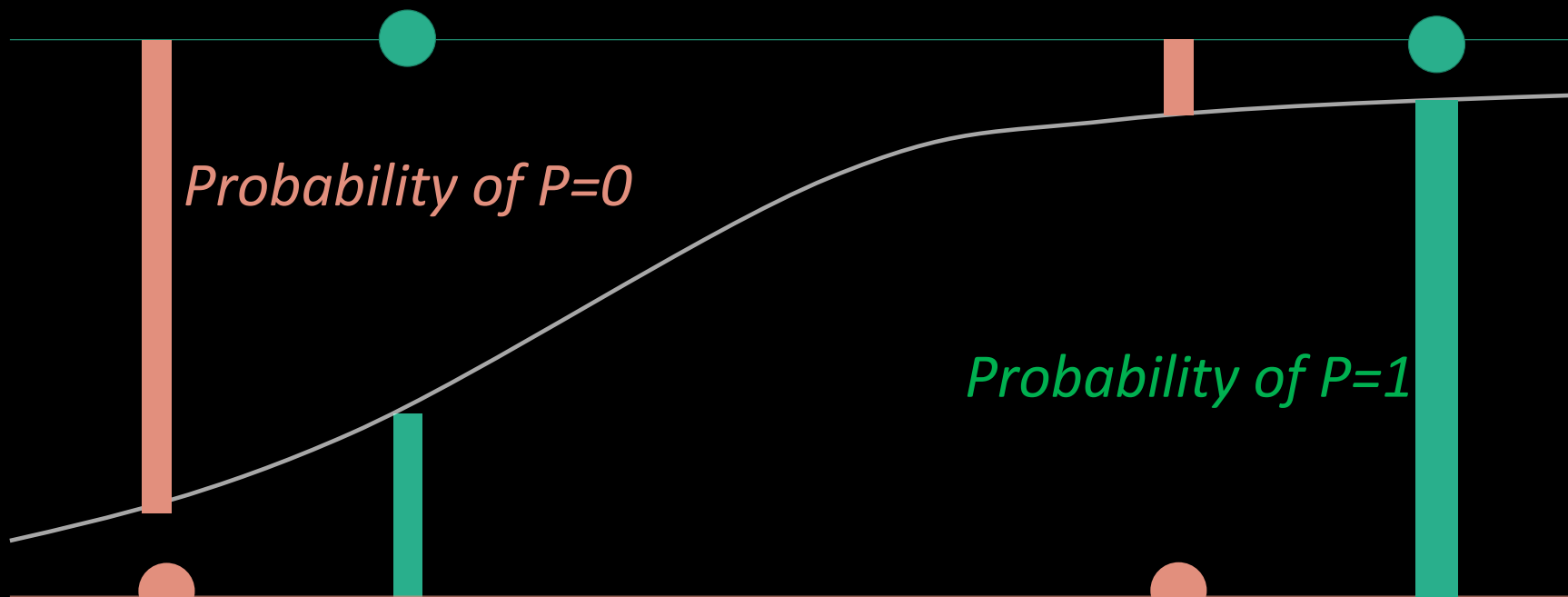
dog



How right are we?

wolf

$P=1$



Probability of $P=0$

Probability of $P=1$

$P=0$

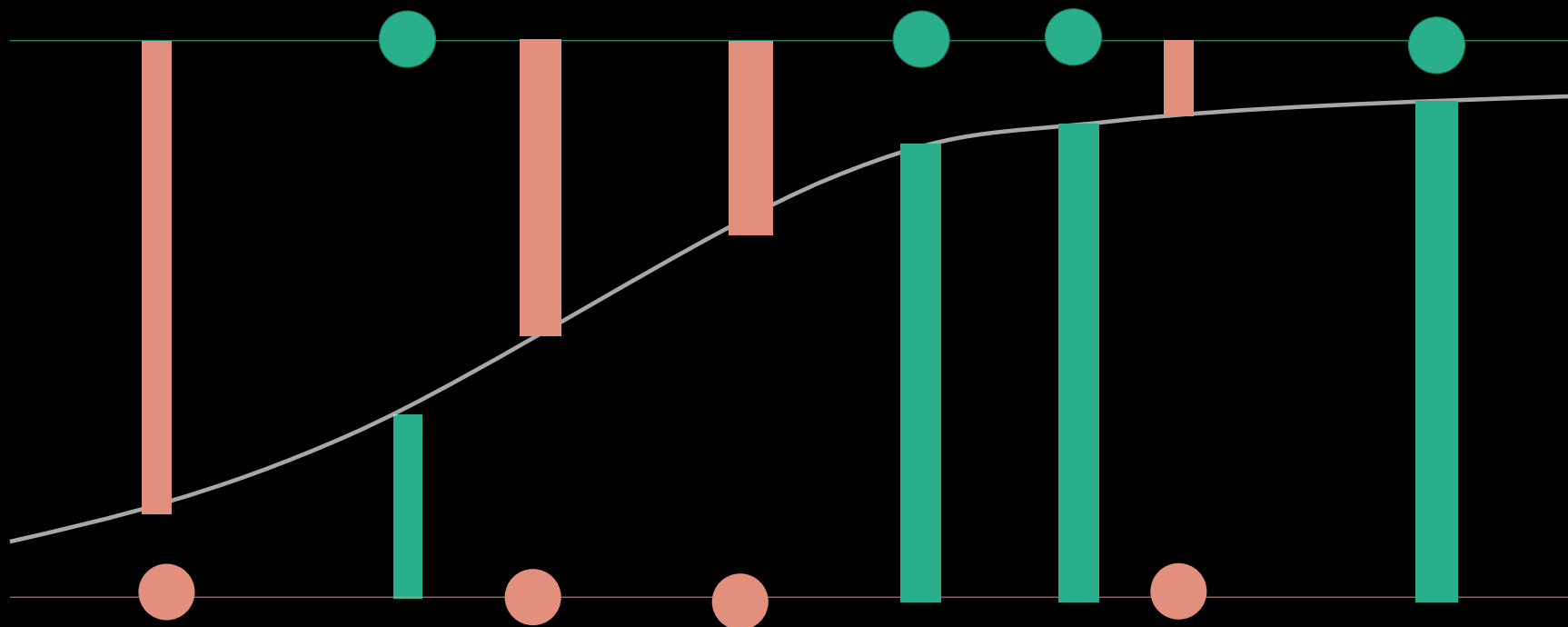
dog

How right we are?

Probability of all the cases = $p_0^ p_1^* p_2^* p_3^* p_4$*

$P=1$

$P=0$



Probability of all the cases

$$(+)\ p_0 * p_1 * p_2 * \dots \dots$$

Remember log(Prob.)

$$\prod_n P(y_n | x_n; \theta)$$

VS

$$(-)\ p_0 * p_1 * p_2 * \dots \dots$$

$$\sum_i \log P(y_n | x_n; \theta)$$

Binary Cross Entropy (BCE) loss

$$y = 1$$

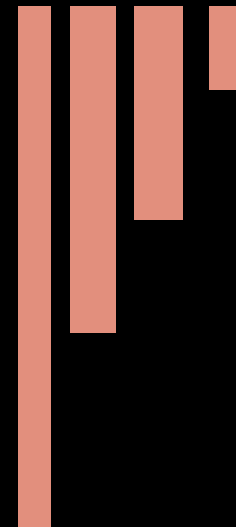
$$y = 0$$

$$BCE = - \sum_n \overset{1}{y} \cdot \log(p(\hat{y})) - \sum_n \overset{1}{(1 - y)} \cdot \log(p(1 - \hat{y}))$$

*All the probs
predicted $y = 1$
correctly*

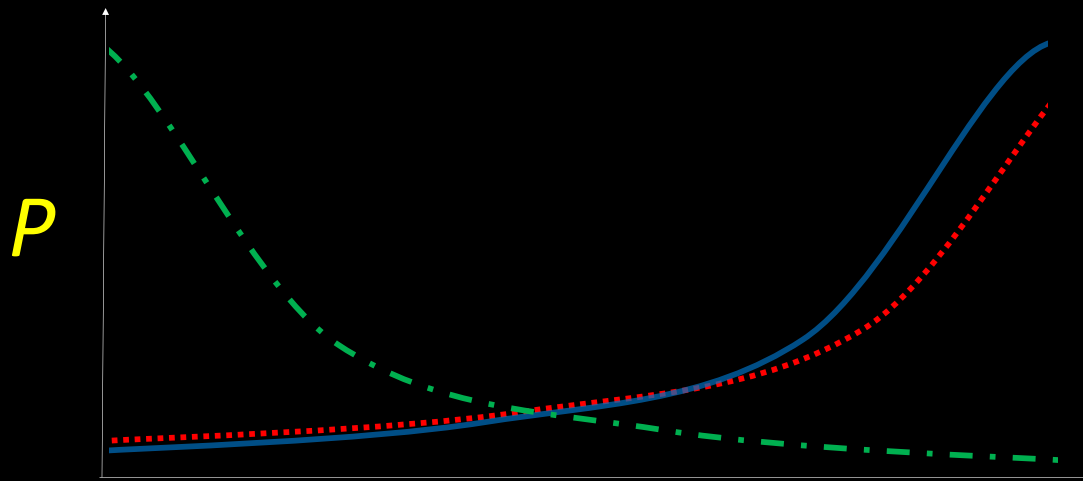


*All the probs
predicted $y = 0$
correctly*



Again, this is based on Maximum Likelihood Estimation (MLE)

$$\theta_{ML} = \operatorname{argmax}_{\theta} P(y|X; \theta)$$



Q , Real Probability

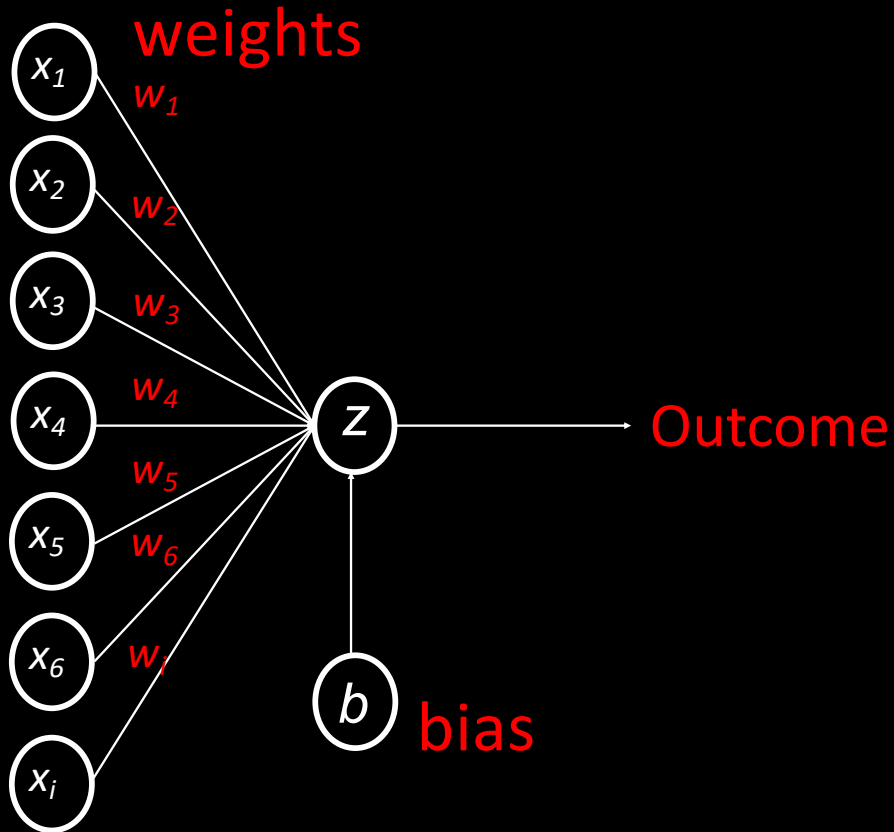
$P(y|X; \theta)$, good prediction

$P(y|X; \theta)$, bad prediction

Cross Entropy:

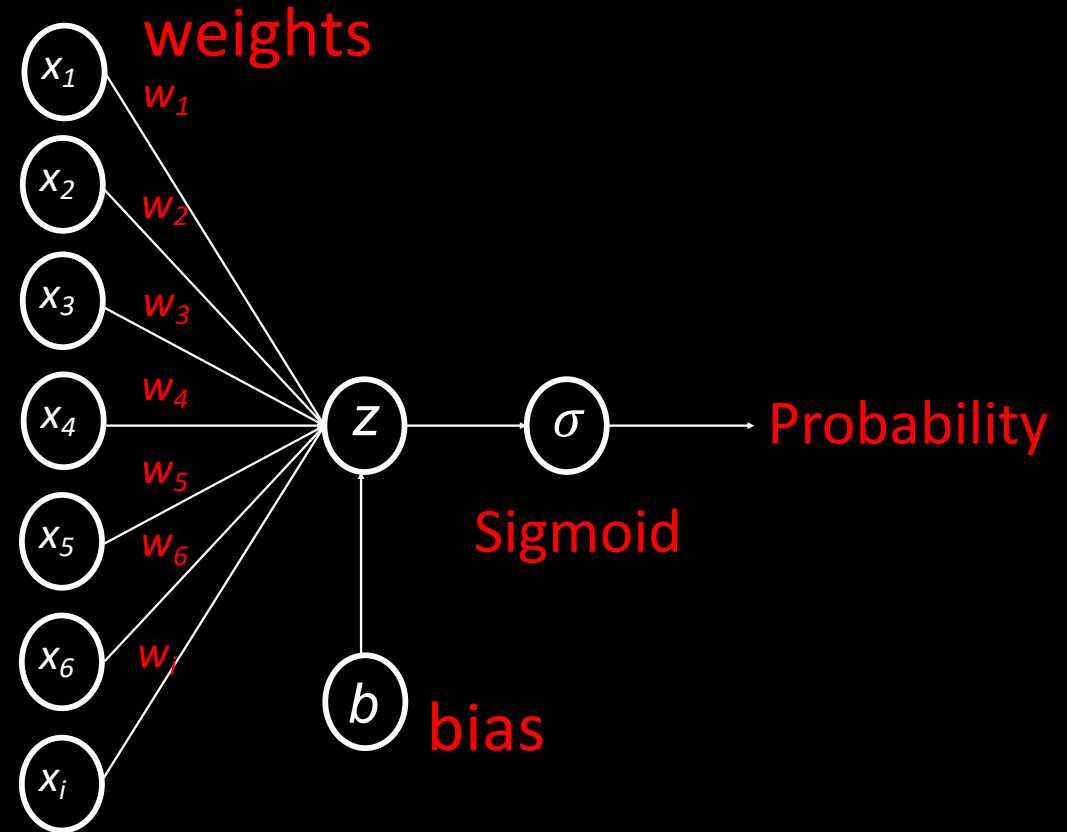
how different they are?

Summary: Regression



Loss: Mean Square error

Classification (logistic regression)



Loss: Cross Entropy